

Spending, Care Access, and Quality following Private Equity and Other For-Profit Acquisitions of Mental Health and Substance Use Treatment Facilities in the US

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Abstract

Importance: Private equity (PE) and other for-profit firms have fueled a growing wave of healthcare consolidation through acquisitions. Behavioral healthcare has been a target, as serial acquisitions of its smaller, independent providers can generate market power while avoiding antitrust scrutiny. Understanding implications for vulnerable populations such as Medicare beneficiaries is particularly important, as over 20% of Medicare beneficiaries are affected by a mental illness or substance use disorder (SUD).

Objective: To determine whether for-profit acquisitions of behavioral healthcare facilities were associated with changes in Medicare beneficiaries' care access and outcomes.

Design: Cohort study using Medicare fee-for-service claims (2014-2021) and difference-in-differences models comparing acquired and non-acquired facilities before and after acquisition, overall and by acquirer type (private equity, other for-profit).

Setting: Specialty behavioral health treatment facilities with Medicare patients.

Participants: 34 PE acquired and 59 other for-profit acquired facilities matched to 829 never-acquired comparator facilities, serving ~2.5 million beneficiaries between 2014 and 2021. We required acquired facilities to be observed at least one-year pre-and-post acquisition, non-acquired to be in counties with at least one acquisition, and beneficiaries to be at least 18 years old with calendar-year continuous enrollment.

Main Outcomes and Measures: Beneficiary volume; total and per-beneficiary spending; and quality measures such as follow-up after an acute care event, care continuity, and medication adherence.

Results: There were no significant associations between PE acquisition and measures of spending, utilization, or quality. Acquisition by other for-profit firms was associated with significant increases in annual beneficiary spending 53.3% (95% CI: 44.7%, 61.9%), volume 31.5% (95% CI: 21.1%, 42.0%), and spending per beneficiary 16.1% (95% CI: 2.3%, 30.0%) compared to matched never-acquired facilities pre-post-acquisition.

Conclusions and Relevance: PE acquisition was associated with small, insignificant increases in beneficiary spending and volume while other for-profit acquisitions were associated with large, significant increases, highlighting differences that may help policymakers anticipate whether an acquisition is likely to increase Medicare volume and service intensity.

Introduction

For-profit investors, including private equity firms (PE) and other corporate entities, such as health systems and hospitals, have played a growing role in the consolidation of mental health and substance use care through acquisitions. Historically, the medical community's view of behavioral health as distinct from physical health led to a fragmented "system" of largely independent specialty providers operating outside the broader medical system and characterized by lower reimbursement rates and less generous insurance coverage compared to other services.^{1,2} More recently, efforts to mitigate these issues have contributed to the expansion of coverage and helped to destigmatize behavioral healthcare, driving increased demand for services and an influx of for-profit investors, who acquired nearly 3,000 behavioral health treatment facilities between 2010-2021.^{3,4,5,6,7} These facilities deliver a large share of behavioral healthcare in the US, raising questions regarding the implications for treatment access and health outcomes.⁶

The share of acquisitions by private equity (PE) firms – a type of for-profit investor – has also grown substantially in the recent past, raising further questions over how these implications may differ among facilities acquired by PE versus other for-profit firms.⁸ PE firms typically employ a short-term, capital-intensive, and highly leveraged "platform-and-rollup" strategy in which an initial "platform" acquisition is followed by a series of smaller acquisitions that are "rolled-up" to the platform, before leveraging economies of scale, cost-cutting techniques, and managerial interventions to generate profits through resale, usually to a larger PE firm that repeats the process at a greater scale.⁹ This approach is well-suited to fragmented markets like behavioral healthcare, where market power can be amassed relatively quickly without any one transaction exceeding the federal antitrust threshold of ~\$130 million.¹⁰ Other for-profit investors, who are often healthcare facilities themselves, may target investments that align with pre-existing clinical operations and corporate structures, with less focus on short-term resale value and debt-financed transactions.

Medicare enrollment rates have accelerated since 2010 when large swaths of the baby boomer generation started becoming eligible, incentivizing for-profit entry into these growing markets.^{11,12,13} At the same time, nearly a third of Medicare beneficiaries aged 65+ and two-thirds of younger adults enrolled in Medicare due to disability have a mental illness or substance use disorder (SUD), yet more than half of affected beneficiaries receive no treatment due to limited provider availability, high out-of-pocket costs, stigma, and other challenges.^{12,14,15,16,17,18,19,20} Crucially, it remains unclear whether this increased for-profit investment and ownership will mitigate, worsen, or have no effect on these issues. In other areas of healthcare, acquisition has largely been associated with increased spending and utilization, but these patterns are difficult to generalize to behavioral healthcare, given its unique financial, clinical, and regulatory environments.^{21,22,23} To address this gap, we sought to answer two core research questions: (1) what happens to spending, utilization, and treatment quality at behavioral health treatment facilities after for-profit acquisition, and (2) what are the differences between those acquired by PE versus other for-profit firms?

Methods

This study followed Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guidelines and was deemed exempt by the Johns Hopkins University Institutional Review Board.

Data Sources

We used data on specialty behavioral health treatment facilities collected by the Substance Abuse and Mental Health Services Administration (SAMHSA), including the National Survey of Substance Abuse Treatment Services (N-SSATS) and the National Mental Health Services Survey (N-MHSS) facility directories, and we identified acquisitions using a structured web-search and matching algorithm described in prior work and summarized in eTable 1.^{7,24} We identified acquisitions in 2014-2021 fifty-state Medicare fee-for-service claims by matching facility addresses to the National Plan and Provider Enumeration System – the national

directory of all active providers in the US – which allowed us to recover organization-level National Provider Identifiers (NPI) that were then linked to Medicare claims. 30.4% of all identified facilities were matched to claims, including 30.5% of PE, 35.0% of other for-profit, and 29.8% of never acquired facilities. These data were then rolled-up to the facility-year level. We required beneficiaries to have calendar-year continuous enrollment and be at least 18 years of age, given behavioral health conditions account for a substantial share of Medicare enrollment through disability.²⁵ Dually eligible beneficiaries were included, though we do not capture Medicaid claims in this study. We also required that acquired facilities have at least one pre and one post-acquisition observation, and that never-acquired comparator facilities be located in a county where at least one acquisition had occurs. Each acquired facility was matched to 30 non-acquired comparators via exact matching on census region and year and nearest-neighbor matching on pre-acquisition beneficiary volume and the proportion receiving inpatient services and with SUD, respectively. The resulting analytic sample was comprised of 93 acquired (59 PE, 34 other for-profit) and 846 never-acquired facilities. No facility was acquired more than once during the study period. Acquisitions by year are described in eTable 2.

Measures

-Independent Variables

We created two variables to identify acquisitions: (1) a variable equal to one beginning the year a facility was acquired, and zero otherwise, and (2) a variable indicating whether the acquisition was by PE or another for-profit investor. Facility characteristic measures included census region, whether the facility had any claims for inpatient or outpatient services, the percentage of unique beneficiaries with any mental health condition, a serious mental illness (SMI; defined as schizophrenia or bipolar disorder), a substance use disorder (SUD), alcohol use disorder (AUD), or an opioid use disorder (OUD); and ownership type (for-profit, nonprofit, government). We also measured shares of beneficiaries by dual eligibility status, race/ethnicity,

and age. Facility acuity was assessed by beneficiary hierarchical condition category scores (HCC).²⁶ Relevant ICD-10 codes can be found in eTable 3.

Outcome Variables

-Spending

Spending was measured both overall and by service type (outpatient, inpatient) as total annual beneficiary expenditure on treatment services provided by a given facility. Because prices are administratively set in Traditional Medicare, we also measured spending per beneficiary, to disentangle changes in service volume and intensity from underlying changes in beneficiary volume. Observations within the top and bottom 2.5% of total spending were winsorized to account for extreme outliers common in claims expenditure data.

-Utilization

Utilization was primarily measured as the annual volume of unique beneficiaries using services provided by a given facility, both overall and by service type. To account for differences in utilization intensity and regularity, we constructed a measure of “attributed” beneficiary volume, as the subset of unique beneficiaries receiving the plurality of their treatment at a given facility. We also measured use of Medications for Opioid Use Disorder (MOUD).

-Quality

We included several measures of treatment quality including the percentage of beneficiaries who: had an emergency department (ED) visit with a primary mental health or substance use-related diagnosis, received outpatient follow-up care within at least seven or at least thirty days of a mental health or substance use-related ED visit or inpatient stay, had at least 3 months of continuous outpatient mental health or substance use treatment use, and who received antipsychotic medication with an active psychotic disorder diagnosis. Among beneficiaries that received psychotropic medication, we also measured their total months with a medication possession ratio above 0.80. National drug and procedure codes used to construct these measures came from the Mental Health Research Network.²⁷

Statistical Analysis

We estimated stacked difference-in-differences models comparing outcomes between for-profit acquired and matched never-acquired facilities, pre-post acquisition.²⁸ Specifically, acquired facilities were grouped into cohorts by year of acquisition and paired with a set of matched never-acquired comparator facilities re-indexed from calendar to acquisition-relative years. Cohorts were then “stacked” around time zero to form a dataset conducive to standard difference-in-differences estimation, with cohort-specific facility and year fixed-effects absorbing calendar-year and facility-specific variation within cohorts. Models interacted with acquirer-type were also estimated to assess post-acquisition differences between PE and other for-profit acquisitions. Models were specified with no covariate adjustment beyond the requisite fixed-effects, as additional adjustment can limit external validity and introduce bias or heteroskedasticity.^{29,30} Independent variables were used peripherally in sensitivity testing, described below, and to construct and describe our matched analytic dataset. We used ordinary least squares (OLS) for estimation of all outcomes except spending and volume, which were estimated via poisson regression. All models were weighted by beneficiary volume, except for those in which volume was the outcome being estimated. Standard errors were clustered at the facility level. If the standard assumptions are met, these models estimate the independent relationship between acquisition and outcomes. Most notably, outcomes would need to have evolved similarly over time in the absence of acquisition.

Several sub-analyses were carried out to support this assumption of parallel trends and to contextualize our core difference-in-differences analysis. First, we computed mean facility characteristics and outcomes across PE, other for-profit, and never-acquired facilities. Bonferroni-adjusted chi-squared tests were used to test differences across these groups both pre and post acquisition. Second, we plotted mean facility outcomes to visually assess whether these groups exhibited parallel trends prior to acquisition and the possibility that they would have continued along this trajectory had no acquisition occurred. Finally, we estimated event

studies to statistically assess the parallel trends assumption after fixed-effects adjustment, which we further scrutinized by applying the Honest Difference-in-Differences framework.³¹ All analyses were conducted in R version 4.5.1.

Sensitivity Tests

We tested the robustness of significant findings in several ways. First, we tested for the influence of highly specialized facilities (opioid treatment programs (OTPs), memory/Alzheimer's care, and eating disorder treatment) by dropping them from the sample. Second, we conducted additional exact matching on facility ownership to rule out confounding from underlying operational differences. Third, we estimated models with additional cohort-indexed state fixed-effects to account for factors like differences in state policy environments and the geographic penetration of PE and other for-profit acquirers. The geographic distribution of PE acquired, and other for-profit acquired facilities was also described for context. Fourth, we tested robustness to Bonferroni adjustment, given our examination of multiple outcomes. Fifth, we tested for contamination of our comparator group by estimating difference-in-differences models comparing outcomes at never-acquired facilities pre-post an acquisition occurring in their county to other never-acquired facilities located in counties where an acquisition had yet to take place. Sixth, we tested the robustness of our model specification by comparing our main findings with estimates from unweighted models, models estimated via OLS and estimates from a range of alternative difference-in-differences estimators.^{32,33,34} Seventh, we tested for compositional changes that may obscure or influence trends in outcomes by estimating event studies of acquisition on average beneficiary HCC risk score and case mix. Eighth, we tested for confounding with the expansion of Medicare coverage for OUD treatment services in January, 2020 by estimating models on a truncated dataset spanning 2014-2019. Finally, we assessed the influence of beneficiaries receiving care from multiple sites by estimating models assigning beneficiaries to the facility where they received the plurality of their care.

Results

Table 1 describes key characteristics of and average outcomes at acquired facilities before and after acquisition and their matched never-acquired comparators. Facilities were evenly geographically distributed, though PE acquisitions were slightly more concentrated in the South (26/44% of facilities) and Midwest (22/37%). Pre-acquisition, 54 PE (92%), 24 other for-profit (71%), and 286 never-acquired facilities (28%) were for-profit owned. There were no changes in never-acquired facility ownership during the study period. 15 PE (25%), 6 other for-profit (18%), and 261 never-acquired comparators (31%) offered inpatient services, the remainder only offered outpatient care. The percentage of beneficiaries with any mental health condition before acquisition was 92.2% among PE and 91.5% among other for-profit acquired facilities growing slightly to 94.0% among PE and 95.1% among other for-profit acquired facilities post-acquisition. Shares of beneficiaries with SUD followed a similar pattern (PE: 24.8% pre, 23.2% post; other for-profit: pre 24.1%, post: 24.5%). 96.0% of beneficiaries at matched never-acquired facilities had a mental health diagnosis, and 30.8% had SUD. The rate of mental health-related ED visits among beneficiaries at PE acquired facilities was 9.5% pre-acquisition (other for-profit: 18.8%) and 15.4% post (other for-profit: 17.3%), and 15.2% (other for-profit: 12.4%) of those beneficiaries received outpatient follow-up care within 7 days pre-acquisition, and 14.8% (other for-profit: 19.5%) did post-acquisition. Among matched never-acquired facilities, the mental health-related ED visit rate was 16.5% and 14.8% received follow-up outpatient care within 7 days. eTable 4 contains descriptive statistics and Bonferroni-adjusted hypothesis tests of the comprehensive set of measured facility characteristics and outcomes; most measures were statistically similar across facilities in the pre-acquisition period. Exceptions included measures of demographic and case mix and shares of dually eligible beneficiaries.

Figure 1 shows that in the three years preceding acquisition, total beneficiary spending decreased from an average of \$237,380 to \$197,244 at PE-acquired facilities, from \$82,786 to \$80,184 at other for-profit acquired facilities, and from \$164,227 to \$155,435 at never-acquired

facilities. Beneficiary volume at PE acquired facilities increased slightly from 40 to 48 beneficiaries and decreased among other for-profit and never-acquired facilities, from 116 to 100 beneficiaries, and from 63 to 61 beneficiaries, respectively. Spending per beneficiary decreased from \$4,205 to \$3,204 at PE acquired facilities, from \$1,690 to \$1,630 at other for-profit acquired facilities, and from \$2,797 to \$2,723 at never-acquired facilities. There were substantial post-acquisition increases among other for-profit acquired facilities in total spending, from \$102,206 to \$152,136, and in spending per beneficiary, from \$1,865 to \$2,897. eFigure 1 decomposes these trends into inpatient and outpatient spending and volume. Most notably, spending on inpatient services grew from 42% (\$42,762) to 70% (\$106,550) of all spending at other-for profit acquired facilities over the three years post-acquisition.

Event studies show that in the pre-acquisition period, outcomes did not significantly differ between acquired and non-acquired facilities, regardless of acquirer type (Figure 2). In contrast, there were significant post-acquisition increases in beneficiary spending and volume among other-for profit acquired facilities. Relative to never-acquired facilities, beneficiary spending among other for-profit acquired facilities increased by 56% (95% CI: 43%,70%) in the first-year post-acquisition, 54% (95% CI: 46%,62%) in the second, and 63% (95% CI: 53%,72%) in the third. Beneficiary volume followed a similar pattern [19% (95% CI: 9%,29%), 24% (95% CI: 9%,34%), 36% (95% CI: 15%,58%)]. There were also estimated increases in spending per beneficiary, though not all significant (36% [95% CI: 7%,66%], 34% [95% CI: 12%,57%], 16% [95% CI: -32%,64%]). Changes among PE acquired facilities were smaller and not statistically significant. Underlying these results, we estimated significant post-acquisition increases in inpatient spending and utilization among other for-profit acquired facilities (eFigure 2), though smaller sample sizes contributed to less precision than overall estimates, making it more difficult to rule out pre-acquisition differences in these sub-analyses.

Figure 3 depicts key stacked difference-in-differences estimates. Compared to non-acquired facilities, there were no significant changes in spending, volume, and spending per

beneficiaries at facilities acquired by PE. In contrast, among other for-profit acquired facilities compared to those that were never acquired, beneficiary spending increased by a significant 53.3% (95% CI: 44.7%,61.9%), spending per beneficiary by 16.1% (95% CI: 2.3%,30.0%), and volume by 31.5% (95% CI: 21.1%,42.0%), after versus before acquisition. For reference, spending on services provided by other for-profit acquired facilities in the average pre-acquisition year was \$74,682 overall and \$1,637 per beneficiary, and total volume was around 76 unique beneficiaries per year. These estimates were comprised primarily of significant increases in average inpatient spending and volume among other for-profit acquired facilities and significant average increases in outpatient volume at both PE and other for-profit acquired facilities relative to matched never-acquired facilities, pre-post-acquisition (eFigure 3). Relative to changes among PE-acquired compared to matched never-acquired facilities, other-for profit acquired facilities saw a 58.0% (95% CI: 27.5%, 63.9%) greater increase in spending, an 18.3% (95% CI: 0.3%, 33.2%) greater increase in spending per beneficiary, and no significant differences in beneficiary volume (eFigure 4).

Estimates across all measures of quality were either not statistically significant, were not estimable due to small sample sizes, or there was not strong evidence to support the assumption of parallel trends. Unadjusted trends and event studies with their corresponding difference-in-differences estimates for these measures are reported in the online supplement (eFigures 5-11).

-Sensitivity Testing Results

Sensitivity testing broadly supported the magnitude, direction, and significance of our main estimates of beneficiary spending, volume, and spending per beneficiary. Since spending per beneficiary was already the least precisely estimated outcome in main analyses, it was almost exclusively the only measure to show sensitivity to testing. eFigure 12 shows robustness under the Honest Difference-in-Differences framework, with marginally less confidence in estimates of spending per beneficiary. eFigure 13 shows that significant findings were stable in models

omitting highly specialized facilities (OTPs, Alzheimer's/memory care, and eating disorder treatment facilities). eFigure 14 shows that estimates were robust to additional matching on pre-acquisition facility ownership. eFigures 15 and 16 show that main findings were robust to state fixed-effects specifications that absorb all within-state variation, and corroborating these findings, eFigure 17 shows that acquisitions did not often occur within the same state. eFigure 18 shows that findings for spending and spending per beneficiary were robust to multiple hypothesis adjustment. eFigures 19, 20, and 21 show that models estimated via weighted least squares (WLS), ordinary least squares (OLS), and unweighted poisson regression produced similar, but somewhat less precise estimates than main models. A similar pattern held for models estimated via alternative difference-in-differences estimators, shown in eFigures 22, 23, and 24 likely because most of these estimators do not support poisson regression so were estimated via OLS. eTable 5 shows that measured characteristics of never-acquired facilities did not change following the acquisition of a nearby facility, implying that the comparator group was not contaminated. eFigure 25 shows that models run on data limited to the years prior to the COVID-19 pandemic and expanded OUD treatment coverage in Medicare (2014-2019) were consistent with main findings, though less precise, likely due to reduced statistical power. eFigures 26-28 show that there was no change in average beneficiary HCC risk scores, the proportion that were dually enrolled in Medicaid, nor in patterns of utilization, suggesting that findings were not driven or obscured by compositional changes beneficiary in acuity or treatment need following acquisition. eFigure 29 shows that estimates did not depend on whether we assigned beneficiaries to the facility where they received the plurality of their care.

Discussion

In this quasi-experimental study of fee-for-service Medicare beneficiaries that received services from specialty behavioral health treatment facilities between 2014 and 2021, we found that acquisition by a for-profit firm was associated with increased total beneficiary spending, volume, and spending per-beneficiary, with other for-profit (non-PE) acquisitions explaining the

bulk of this variation. We found no association between acquisition and measured treatment quality; however, these null findings should be interpreted cautiously given limited statistical power for these outcomes.

Though insignificant, PE acquisition was associated with a downward trend in average spending and volume among facilities that started with higher pre-acquisition levels. On the other hand, other for profit-acquisition was associated with a significant upward trend among facilities that started with low pre-acquisition levels of spending and volume. Together, these trends effectively brought outlier facilities towards market averages.

This observed convergence may be an artifact of differences in PE and non-PE investment strategy. PE firms' relatively short investment horizons and role as corporate owners, rather than clinical operators, may incentivize acquisitions of facilities that already perform well financially, where profits can be generated quickly through the economies of scale afforded by consolidation and through cost control measures like staffing and administrative consolidation.^{6,35,36} In contrast, other for-profit acquirers, which are often hospitals or health systems, could leverage existing infrastructure, such as referral networks and contracts with providers to generate profits, targeting facilities with longer-term growth potential, and bringing them closer to market averages that, as healthcare entities large enough to conduct acquisitions, they likely help set.^{37,38} Whether firms increase hiring to accommodate this increased patient volume remains an open question; if they do not, profit margins would widen but at the potential cost of overburdened staff and diminished treatment quality.

Limitations

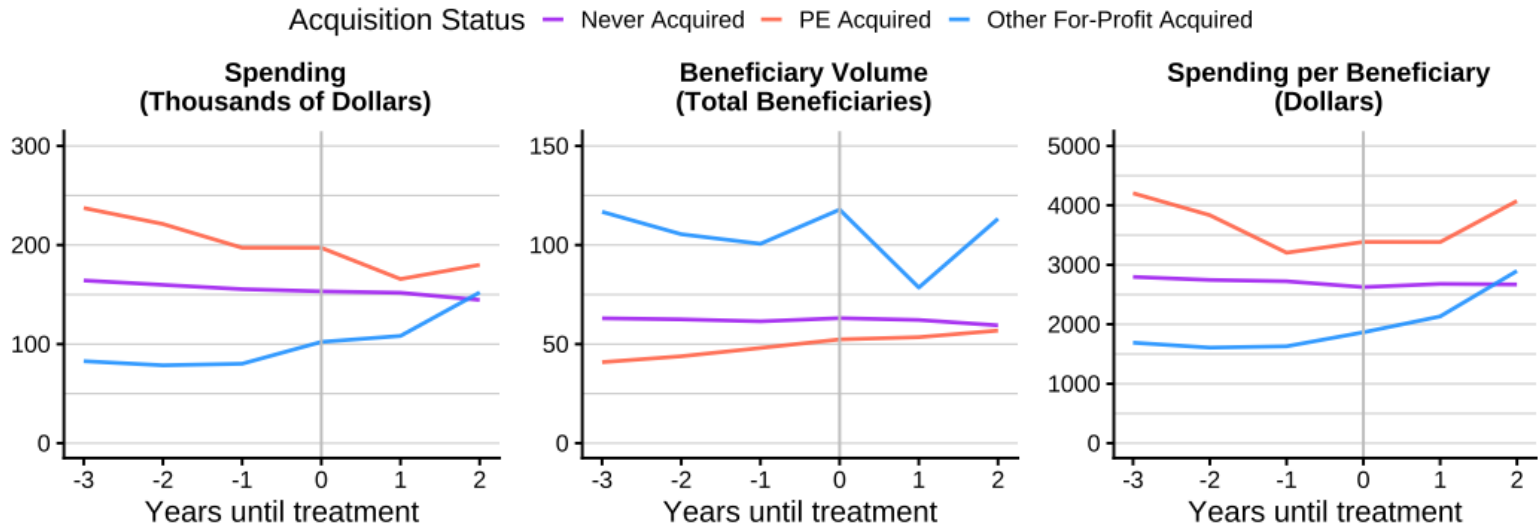
This study has several limitations. First, the number of acquired behavioral health treatment facilities billing Traditional Medicare is relatively small. While the sample was sufficient to detect meaningful changes, subgroup analyses and assessments of measured care quality, which are often observed infrequently relative to measures of overall utilization and spending, were limited by smaller numbers. Second, this study is limited to care provided by

dedicated behavioral health treatment facilities, which does not include other key treatment settings, like primary care, so the sample may reflect individuals with more complex or serious treatment needs. Third, it is possible that some for-profit acquisitions were omitted or misclassified. That said, we used best practice data collection procedures, and our sample is consistent with other, independently conducted research.⁵

Conclusions

In a national sample of fee-for-service Medicare beneficiaries, for-profit behavioral healthcare acquisition was associated with increased beneficiary spending and volume driven primarily by non-PE transactions, with no change in measured treatment quality. By highlighting differences between PE and other for-profit behavioral healthcare acquisitions, findings may help policymakers anticipate whether an acquisition is likely to increase Medicare volume and service intensity.

Figure 1. Facility Average Total Beneficiary Spending, Volume, and Spending Per Beneficiary in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Behavioral Health Treatment Facilities

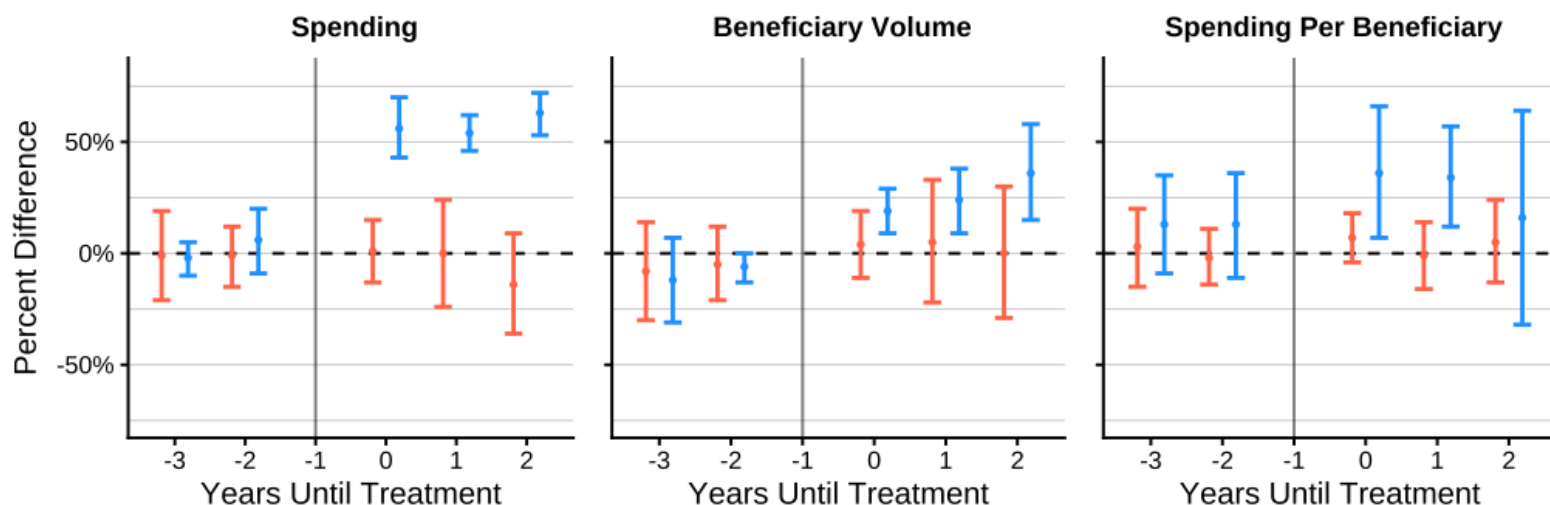


Notes: Averages were weighted by calendar year observations, but no further covariate adjustment was conducted. Acquired facilities were matched to never-acquired comparators using exact matching on census region and year and nearest-neighbor matching on pre-acquisition beneficiary volume and the proportion receiving inpatient services and with SUD, respectively.

Figure 2. Event Studies of Acquisition on Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type

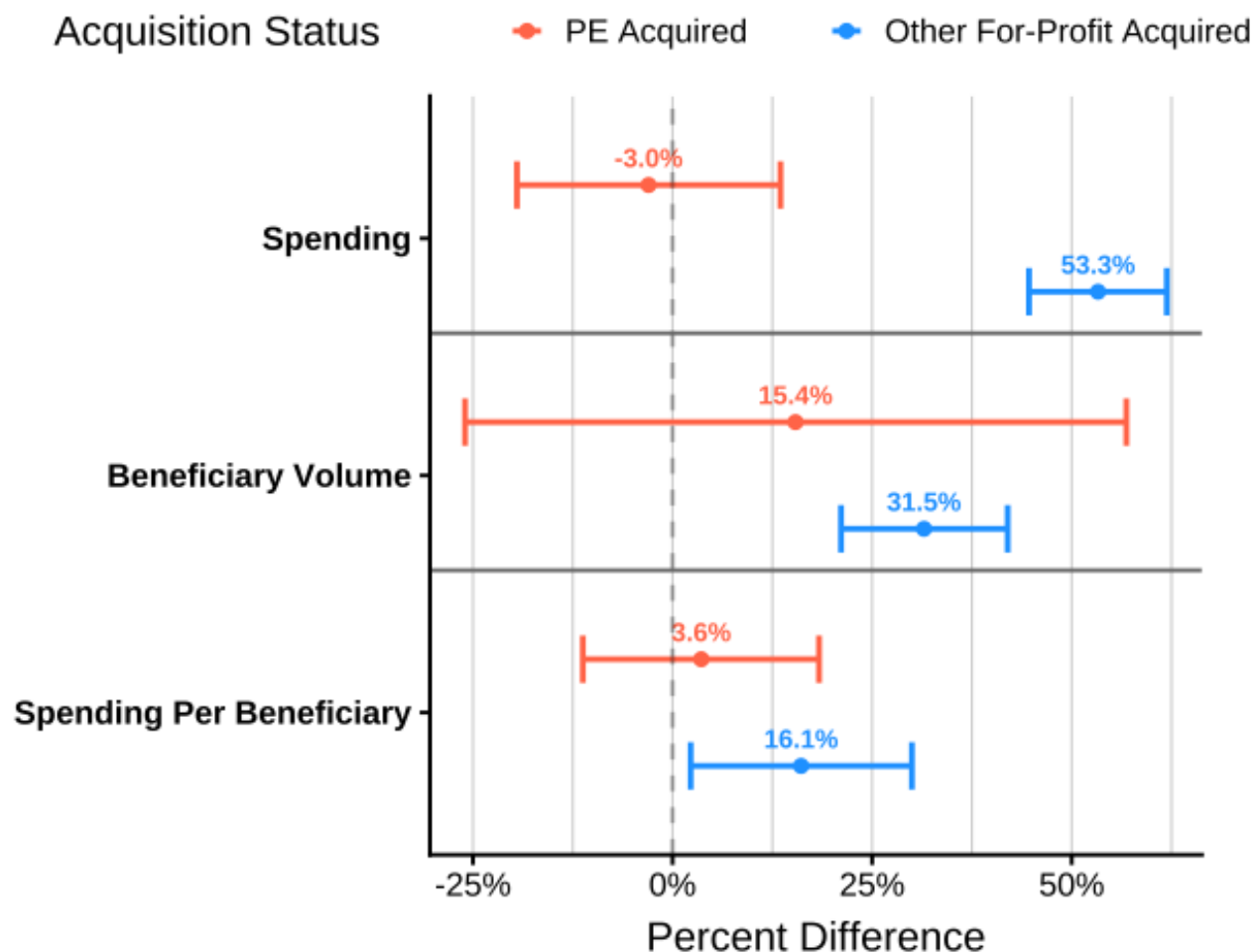
Notes: Event study estimates include facility and fixed-effects. Estimates of spending and spending per

Acquisition Status — PE Acquired — Other For-Profit Acquired



beneficiary were weighted by total beneficiary volume. Poisson regression used for estimation. Acquired facilities were matched to never-acquired comparators using exact matching on census region and year and nearest-neighbor matching on pre-acquisition beneficiary volume and the proportion receiving inpatient services and with SUD, respectively.

Figure 3. Average Change in Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type



Notes: Stacked difference-in-differences estimates including facility and year fixed-effects. Estimates of spending and spending per beneficiary were weighted by total beneficiary volume. Poisson regression used for estimation. Acquired facilities were matched to never-acquired comparators using exact matching on census region and year and nearest-neighbor matching on pre-acquisition beneficiary volume and the proportion receiving inpatient services and with SUD, respectively.

Table 1. Facility Average Characteristics by Acquisition Status, Before and After Acquisition

Acquisition Timing		Pre-Acquisition		Post-Acquisition	
Acquisition Status	Never Acquired	PE	Other For-Profit	PE	Other For-Profit
Descriptive Statistics of Facilities					
Total Facilities	N = 829	N = 59	N = 34	N = 59	N = 34
Location					
South	257 (31%)	26 (44%)	7 (21%)	26 (44%)	7 (21%)
West	158 (19%)	6 (10%)	8 (24%)	6 (10%)	8 (24%)
Midwest	276 (33%)	22 (37%)	10 (29%)	22 (37%)	10 (29%)
Northeast	138 (17%)	5 (8.5%)	9 (26%)	5 (8.5%)	9 (26%)
Provides Inpatient Services	261 (31%)	15 (25%)	6 (18%)	15 (25%)	6 (18%)
Case Mix (Beneficiary Diagnoses)					
Mental Health Condition	96.0% (11.3%)	92.2% (11.2%)	91.5% (9.9%)	94.0% (9.7%)	95.1% (6.9%)
Serious Mental Illness	54.7% (24.9%)	37.9% (26.1%)	36.1% (23.0%)	39.2% (25.4%)	39.2% (25.0%)
Substance Use Disorder	30.8% (23.9%)	24.8% (25.3%)	24.1% (25.0%)	23.2% (21.9%)	24.5% (24.5%)
Opioid Use Disorder	12.5% (19.3%)	11.7% (22.1%)	11.0% (19.6%)	8.8% (17.1%)	10.5% (19.8%)
Ownership Type					
Private For-Profit	236 (28%)	54 (92%)	24 (71%)	59 (100%)	34 (100%)
Private Nonprofit	487 (59%)	4 (6.8%)	9 (26%)		
Government	106 (13%)	1 (1.7%)	1 (2.9%)		
HCC Risk Score	1.4 (0.6)	1.3 (0.6)	1.5 (0.7)	1.6 (0.6)	1.8 (0.7)
Share Dually Eligible	59.0% (24.5%)	39.6% (30.3%)	48.2% (29.3%)	38.4% (27.1%)	46.9% (27.8%)
Outcome Measures					
Spending					
Total Spending	\$135,141 (\$240,984)	\$143,312 (\$258,516)	\$74,682 (\$124,223)	\$148,329 (\$244,302)	\$106,195 (\$178,108)
On Outpatient Services	\$20,015 (\$30,724)	\$21,870 (\$32,444)	\$25,502 (\$38,768)	\$34,004 (\$44,240)	\$30,029 (\$39,374)
On Inpatient Services	\$112,562 (\$219,757)	\$108,625 (\$214,722)	\$40,405 (\$112,976)	\$91,161 (\$184,625)	\$52,465 (\$144,169)
Spending per Beneficiary	\$2,748 (\$3,667)	\$2,866 (\$4,152)	\$1,637 (\$2,410)	\$2,911 (\$4,134)	\$2,156 (\$3,039)
On Outpatient Services	\$1,261 (\$2,119)	\$1,498 (\$1,809)	\$1,482 (\$3,232)	\$1,565 (\$2,039)	\$1,510 (\$1,682)
On Inpatient Services	\$2,847 (\$5,179)	\$3,345 (\$5,862)	\$1,393 (\$2,929)	\$3,264 (\$6,445)	\$2,100 (\$4,694)
Utilization					
Total Beneficiary Volume	48.6 (79.2)	38.8 (82.0)	76.0 (153.4)	53.5 (101.1)	106.4 (243.4)
Outpatient Volume	31.6 (58.0)	24.5 (68.5)	50.1 (101.6)	37.6 (81.2)	71.9 (169.2)
Inpatient Volume	25.7 (44.3)	15.0 (23.9)	33.5 (55.4)	31.2 (65.2)	49.5 (87.2)
Attributed Beneficiary Volume	22.9 (46.5)	23.8 (72.4)	42.5 (117.4)	22.3 (52.7)	57.0 (176.7)

MOUD Uptake	5.7% (20.3%)	4.8% (21.8%)	1.8% (5.1%)	6.7% (22.7%)	20.2% (35.0%)
Quality					
Mental Health Related ED Visit	16.5% (24.6%)	9.5% (16.2%)	18.8% (29.7%)	15.4% (24.7%)	17.3% (32.2%)
7-Day Outpatient Follow-up	14.8% (21.5%)	15.2% (26.8%)	12.4% (24.0%)	14.8% (18.6%)	19.5% (27.4%)
30-Day Outpatient Follow-up	37.5% (34.0%)	43.8% (38.1%)	26.6% (29.0%)	41.4% (34.2%)	39.1% (36.5%)
SUD-Related ED Visit	1.5% (5.4%)	0.8% (3.8%)	2.6% (13.3%)	1.0% (3.9%)	4.5% (17.6%)
7 Day Outpatient Follow-up	8.9% (20.3%)	0.0% (0.0%)	0.2% (0.6%)	7.1% (26.7%)	9.1% (16.9%)
30 Day Outpatient Follow-up	16.4% (29.5%)	3.5% (10.4%)	13.0% (35.2%)	7.1% (26.7%)	14.7% (31.6%)
3-Months Continuous Outpatient Treatment (Mental Health)	35.2% (25.5%)	45.8% (28.0%)	37.9% (23.8%)	44.3% (25.7%)	42.8% (29.7%)
3-Months Continuous Outpatient Treatment (SUD)	7.8% (14.9%)	8.4% (20.7%)	5.1% (12.9%)	7.2% (15.7%)	7.8% (19.0%)
Antipsychotic Prescription with Psychotic Disorder Diagnosis	4.6% (11.4%)	6.1% (18.5%)	8.2% (22.3%)	9.7% (15.7%)	3.5% (6.5%)
Months of Consecutive Adherence to Psychotropic Medication	4.3 (3.0)	3.1 (3.0)	4.1 (3.3)	6.1 (3.4)	2.0 (1.0)

Notes: Statistics reflect all specialty behavioral health treatment facilities between 2014-2021. “Attributed Beneficiaries” are those that received the plurality of their care at a given facility, and thus are attributed to that facility, while “Overall Beneficiaries” imposes no restriction on which beneficiaries are counted in a given facility’s measured volume. Acquired facilities were matched to never-acquired comparators using exact matching on census region and year and nearest-neighbor matching on pre-acquisition beneficiary volume and the proportion receiving inpatient services and with SUD, respectively. MOUD is medications for opioid use disorder. ED is emergency department.

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eTable 1. Description of Acquisitions Database

Private equity acquisitions were identified using capital market transactions data from PitchBook, Inc., a commercial dataset validated for and commonly used in healthcare acquisitions research.^{39,40,41,42,43,44} To filter for behavioral health transactions, two coders conducted an independent review of PitchBook deal descriptors for all transactions occurring during the study period. There was 98% agreement between the coders' flags for behavioral health transactions. Because one transaction could affect several facilities, we employed a structured web-search process in which we manually searched for and verified the geographic locations of all individual facilities that were affected by each acquisition. We did this using the official websites of the primary target of acquisition listed in the pitchbook data, the MIT Wayback Machine, and date-restricted searches to ensure accuracy to the timing of the transaction.

eTable 2. Acquisitions by Calendar Year and Acquirer Type

	Acquisition Status		
Year	Overall	PE-Acquired	Other For-Profit Acquired
2015	5	2	3
2016	6	4	2
2017	7	3	4
2018	3	3	0
2019	3	1	2
2020	5	2	3
2021	8	1	7

eTable 3. Table of ICD-10 Codes

Condition	ICD-10 Codes
Mental Health	F20–F69, F90–F99
Substance Use Disorder	F11–F19 (excl. F17)
Alcohol Use Disorder	F10
SMI	F20–F31 (excl. F31.8)
Schizophrenia	F20–F29
Bipolar	F30, F31 (excl. F31.8)

eTable 4. Comprehensive Descriptive Statistics and Hypothesis Testing: Characteristics of and Beneficiary Outcomes at PE, Other For-Profit, and Matched Never Acquired Facilities

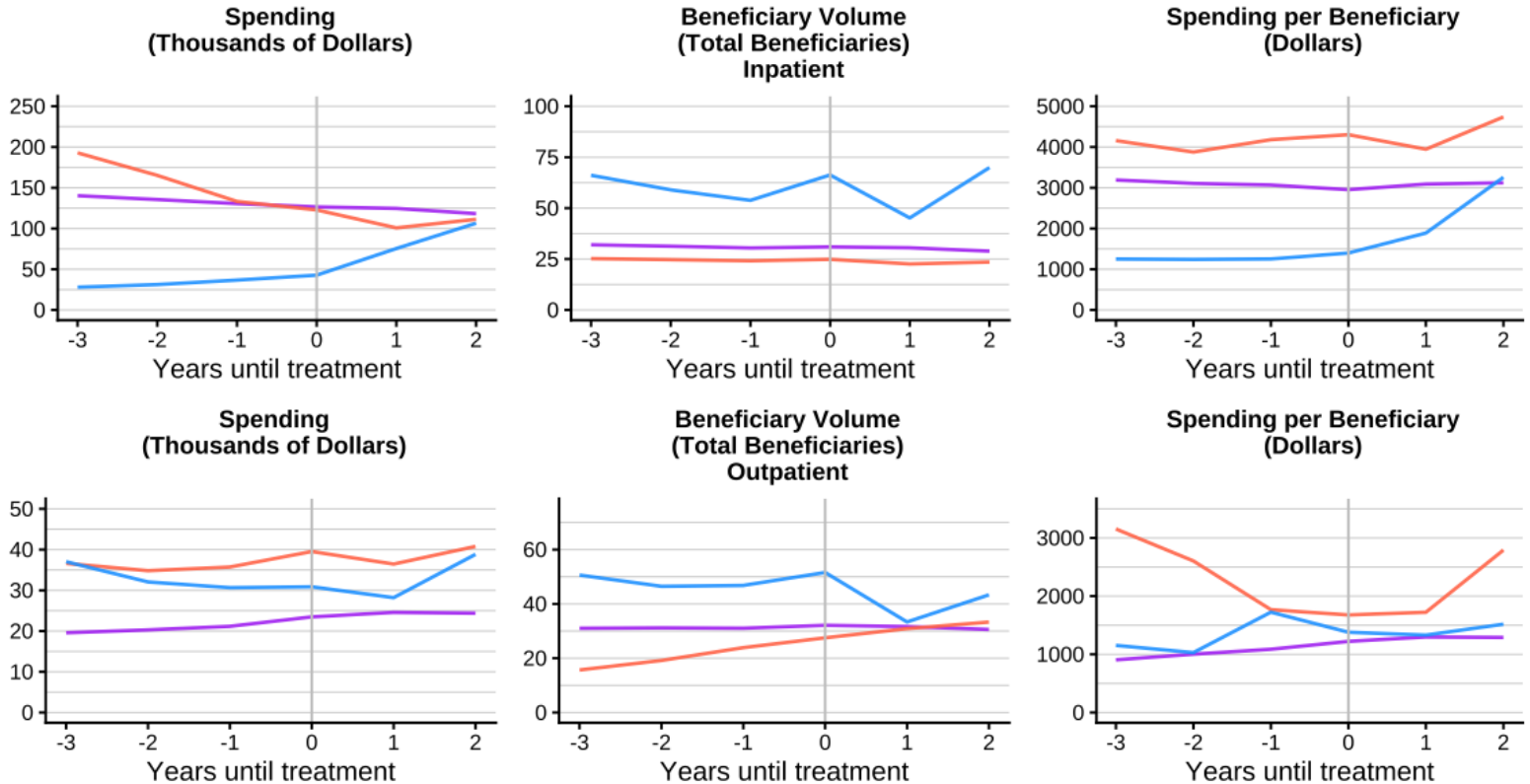
Acquisition Timing		Pre-Acquisition		Post-Acquisition		P-Values			
Acquisition Status	Never Acquired	PE	Other For-Profit	PE	Other For-Profit	Pre-Acquisition: PE vs. Never Acquired	Pre-Acquisition: Other For-Profit vs. Never Acquired	Pre-Acquisition: PE vs. Other For-Profit	Post-Acquisition: PE vs. Other For-Profit
Descriptive Statistics of Facilities									
Total Facilities	N = 829	N = 59	N = 34	N = 59	N = 34				
Location									
South	257 (31%)	26 (44%)	7 (21%)	26 (44%)	7 (21%)	>0.90	>0.90	>0.90	>0.90
West	158 (19%)	6 (10%)	8 (24%)	6 (10%)	8 (24%)	>0.90	>0.90	>0.90	>0.90
Midwest	276 (33%)	22 (37%)	10 (29%)	22 (37%)	10 (29%)	>0.90	>0.90	>0.90	>0.90
Northeast	138 (17%)	5 (8.5%)	9 (26%)	5 (8.5%)	9 (26%)	>0.90	>0.90	>0.90	>0.90
Provides Inpatient Services	261 (31%)	15 (25%)	6 (18%)	15 (25%)	6 (18%)	>0.90	>0.90	>0.90	>0.90
Case Mix (Beneficiary Diagnoses)									
Mental Health Condition	96.0% (11.3%)	92.2% (11.2%)	91.5% (9.9%)	94.0% (9.7%)	95.1% (6.9%)	0.01	<0.01	>0.90	>0.90
Serious Mental Illness	54.7% (24.9%)	37.9% (26.1%)	36.1% (23.0%)	39.2% (25.4%)	39.2% (25.0%)	<0.01	<0.01	>0.90	>0.90
Substance Use Disorder	30.8% (23.9%)	24.8% (25.3%)	24.1% (25.0%)	23.2% (21.9%)	24.5% (24.5%)	0.44	>0.90	>0.90	>0.90
Opioid Use Disorder	12.5% (19.3%)	11.7% (22.1%)	11.0% (19.6%)	8.8% (17.1%)	10.5% (19.8%)	>0.90	>0.90	>0.90	>0.90
Specialized Program									
Opioid Treatment Program (OTP)	42 (5.1%)	1 (1.7%)	1 (2.9%)	1 (1.7%)	1 (2.9%)	>0.90	>0.90	>0.90	>0.90
Memory Care/Alzheimer's/Dementia Care	128 (15%)	11 (19%)	1 (2.9%)	11 (19%)	1 (2.9%)	>0.90	>0.90	>0.90	>0.90
Eating Disorder Treatment Program	109 (13%)	7 (12%)	0 (0%)	7 (12%)	0 (0%)	>0.90	>0.90	>0.90	>0.90
Ownership Type									
Private For-Profit	236 (28%)	54 (92%)	24 (71%)	59 (100%)	34 (100%)	<0.01	<0.01	>0.90	>0.90
Private Nonprofit	487 (59%)	4 (6.8%)	9 (26%)	0 (0%)	0 (0%)	<0.01	0.02	0.76	>0.90
Government	106 (13%)	1 (1.7%)	1 (2.9%)	0 (0%)	0 (0%)	0.39	>0.90	>0.90	>0.90
Demographic Mix									
Asian	1.6% (6.2%)	0.2% (0.7%)	1.6% (3.7%)	0.5% (1.1%)	1.8% (3.5%)	<0.01	>0.90	0.12	>0.90
Black	21.0% (23.0%)	16.0% (18.7%)	7.1% (11.7%)	16.6% (17.0%)	7.5% (11.1%)	>0.90	<0.01	0.96	0.54
White	70.2% (24.6%)	81.0% (18.3%)	83.3% (13.9%)	78.8% (17.4%)	77.5% (20.6%)	0.02	0.09	>0.90	>0.90
Native	0.9% (5.2%)	0.3% (1.4%)	0.8% (2.6%)	0.9% (3.2%)	0.2% (0.7%)	0.35	>0.90	>0.90	>0.90
Unknown	1.5% (3.1%)	0.9% (1.7%)	2.7% (5.8%)	1.7% (2.0%)	9.2% (19.9%)	0.2	>0.90	>0.90	0.26
Hispanic	3.7% (8.5%)	0.9% (2.0%)	2.5% (4.1%)	0.9% (2.6%)	2.4% (3.6%)	<0.01	>0.90	>0.90	0.14
Age									
18-24	1.7% (6.6%)	0.6% (1.0%)	0.7% (2.3%)	0.7% (1.7%)	0.9% (1.8%)	0.06	>0.90	>0.90	>0.90
25-34	9.1% (8.1%)	4.6% (4.7%)	5.0% (6.9%)	6.5% (9.5%)	5.3% (5.3%)	0.06	0.06	>0.90	>0.90
35-44	15.1% (11.8%)	15.8% (18.7%)	12.2% (15.7%)	8.9% (8.4%)	8.9% (7.5%)	0.06	>0.90	>0.90	>0.90
45-54	19.4% (12.1%)	14.2% (9.3%)	16.0% (19.3%)	13.1% (10.0%)	16.7% (11.5%)	0.06	0.64	>0.90	>0.90
55-64	23.1% (12.5%)	20.3% (13.9%)	23.0% (18.6%)	20.8% (17.8%)	17.6% (10.3%)	>0.90	>0.90	>0.90	>0.90
65+	31.2% (24.0%)	44.5% (28.9%)	43.0% (27.6%)	50.0% (27.6%)	50.5% (23.7%)	0.06	0.06	>0.90	>0.90
Enrollment Reason									

Old Age and Survivors Insurance	31.2% (24.0%)	44.5% (28.9%)	43.0% (27.6%)	50.0% (27.6%)	50.6% (23.7%)	0.06	0.06	>0.90	>0.90
Disability Insurance Benefits (DIB)	68.5% (24.1%)	54.9% (28.8%)	56.8% (27.5%)	49.9% (27.6%)	49.0% (23.7%)	0.06	0.06	>0.90	>0.90
End-Stage Renal Disease (ESRD)	0.2% (2.5%)	<0.01% (0.2%)	<0.01% (0.1%)	<0.01% (<0.01%)	0.2% (0.6%)	>0.90	>0.90	>0.90	>0.90
Both ESRD and DIB	0.1% (0.5%)	0.5% (2.9%)	0.2% (1.0%)	0.1% (0.4%)	0.3% (1.2%)	>0.90	>0.90	>0.90	>0.90
HCC Risk Score	1.4 (0.6)	1.3 (0.6)	1.5 (0.7)	1.6 (0.6)	1.8 (0.7)	>0.90	>0.90	>0.90	>0.90
Share Dually Eligible	59.0% (24.5%)	39.6% (30.3%)	48.2% (29.3%)	38.4% (27.1%)	46.9% (27.8%)	<0.01	>0.90	>0.90	>0.90
Outcome Measures									
Spending									
Total Spending	\$135,141 (\$240,984)	\$143,312 (\$258,516)	\$74,682 (\$124,223)	\$148,329 (\$244,302)	\$106,195 (\$178,108)	>0.90	>0.90	>0.90	>0.90
On Outpatient Services	\$20,015 (\$30,724)	\$21,870 (\$32,444)	\$25,502 (\$38,768)	\$34,004 (\$44,240)	\$30,029 (\$39,374)	>0.90	>0.90	>0.90	>0.90
On Inpatient Services	\$112,562 (\$219,757)	\$108,625 (\$214,722)	\$40,405 (\$112,976)	\$91,161 (\$184,625)	\$52,465 (\$144,169)	>0.90	>0.90	>0.90	>0.90
Spending per Beneficiary	\$2,748 (\$3,667)	\$2,866 (\$4,152)	\$1,637 (\$2,410)	\$2,911 (\$4,134)	\$2,156 (\$3,039)	>0.90	>0.90	>0.90	>0.90
On Outpatient Services	\$1,261 (\$2,119)	\$1,498 (\$1,809)	\$1,482 (\$3,232)	\$1,565 (\$2,039)	\$1,510 (\$1,682)	>0.90	>0.90	>0.90	>0.90
On Inpatient Services	\$2,847 (\$5,179)	\$3,345 (\$5,862)	\$1,393 (\$2,929)	\$3,264 (\$6,445)	\$2,100 (\$4,694)	>0.90	>0.90	>0.90	>0.90
Utilization									
Total Beneficiary Volume	48.6 (79.2)	38.8 (82.0)	76.0 (153.4)	53.5 (101.1)	106.4 (243.4)	>0.90	>0.90	>0.90	>0.90
Outpatient Volume	31.6 (58.0)	24.5 (68.5)	50.1 (101.6)	37.6 (81.2)	71.9 (169.2)	>0.90	>0.90	>0.90	>0.90
Inpatient Volume	25.7 (44.3)	15.0 (23.9)	33.5 (55.4)	31.2 (65.2)	49.5 (87.2)	>0.90	>0.90	>0.90	>0.90
Attributed Beneficiary Volume	22.9 (46.5)	23.8 (72.4)	42.5 (117.4)	22.3 (52.7)	57.0 (176.7)	>0.90	>0.90	>0.90	>0.90
MOUD Utilization	5.7% (20.3%)	4.8% (21.8%)	1.8% (5.1%)	6.7% (22.7%)	20.2% (35.0%)	>0.90	>0.90	>0.90	>0.90
Quality									
Mental Health Related ED Visit	16.5% (24.6%)	9.5% (16.2%)	18.8% (29.7%)	15.4% (24.7%)	17.3% (32.2%)	0.08	>0.90	>0.90	>0.90
7-Day Outpatient Follow-up	14.8% (21.5%)	15.2% (26.8%)	12.4% (24.0%)	14.8% (18.6%)	19.5% (27.4%)	>0.90	>0.90	>0.90	>0.90
30-Day Outpatient Follow-up	37.5% (34.0%)	43.8% (38.1%)	26.6% (29.0%)	41.4% (34.2%)	39.1% (36.5%)	>0.90	>0.90	>0.90	>0.90
SUD-Related ED Visit	1.5% (5.4%)	0.8% (3.8%)	2.6% (13.3%)	1.0% (3.9%)	4.5% (17.6%)	0.16	>0.90	>0.90	>0.90
7 Day Outpatient Follow-up	8.9% (20.3%)	0.0% (0.0%)	0.2% (0.6%)	7.1% (26.7%)	9.1% (16.9%)	>0.90	>0.90	>0.90	>0.90
30 Day Outpatient Follow-up	16.4% (29.5%)	3.5% (10.4%)	13.0% (35.2%)	7.1% (26.7%)	14.7% (31.6%)	>0.90	>0.90	>0.90	>0.90
Follow-up After Inpatient Stay									
Within 7 Days of MH-Related Stay	22.1% (27.3%)	38.6% (39.1%)	24.2% (28.8%)	37.7% (37.4%)	33.0% (33.9%)	>0.90	>0.90	>0.90	>0.90
Within 30 Days of MH-Related Stay	41.3% (36.5%)	47.4% (38.2%)	45.5% (33.9%)	53.7% (37.1%)	46.2% (41.3%)	>0.90	>0.90	>0.90	>0.90
Within 7 Days of SUD-Related Stay	9.5% (22.6%)	9.1% (30.2%)	0.5% (1.3%)	7.1% (26.7%)	3.7% (9.8%)	>0.90	>0.90	>0.90	>0.90
Within 30 Days of SUD-Related Stay	16.5% (29.8%)	18.9% (40.2%)	17.9% (37.1%)	15.0% (36.1%)	18.0% (37.4%)	>0.90	>0.90	>0.90	>0.90
3-Months Continuous Outpatient Treatment (Mental Health)	35.2% (25.5%)	45.8% (28.0%)	37.9% (23.8%)	44.3% (25.7%)	42.8% (29.7%)	0.28	>0.90	>0.90	>0.90
3-Months Continuous Outpatient Treatment (SUD)	7.8% (14.9%)	8.4% (20.7%)	5.1% (12.9%)	7.2% (15.7%)	7.8% (19.0%)	0.28	>0.90	>0.90	>0.90
Antipsychotic Prescription with Psychotic Disorder Diagnosis	4.6% (11.4%)	6.1% (18.5%)	8.2% (22.3%)	9.7% (15.7%)	3.5% (6.5%)	>0.90	>0.90	>0.90	>0.90

Months of Consecutive Adherence to Psychotropic Medication	4.3 (3.0)	3.1 (3.0)	4.1 (3.3)	6.1 (3.4)	2.0 (1.0)	>0.90	>0.90	>0.90	>0.90
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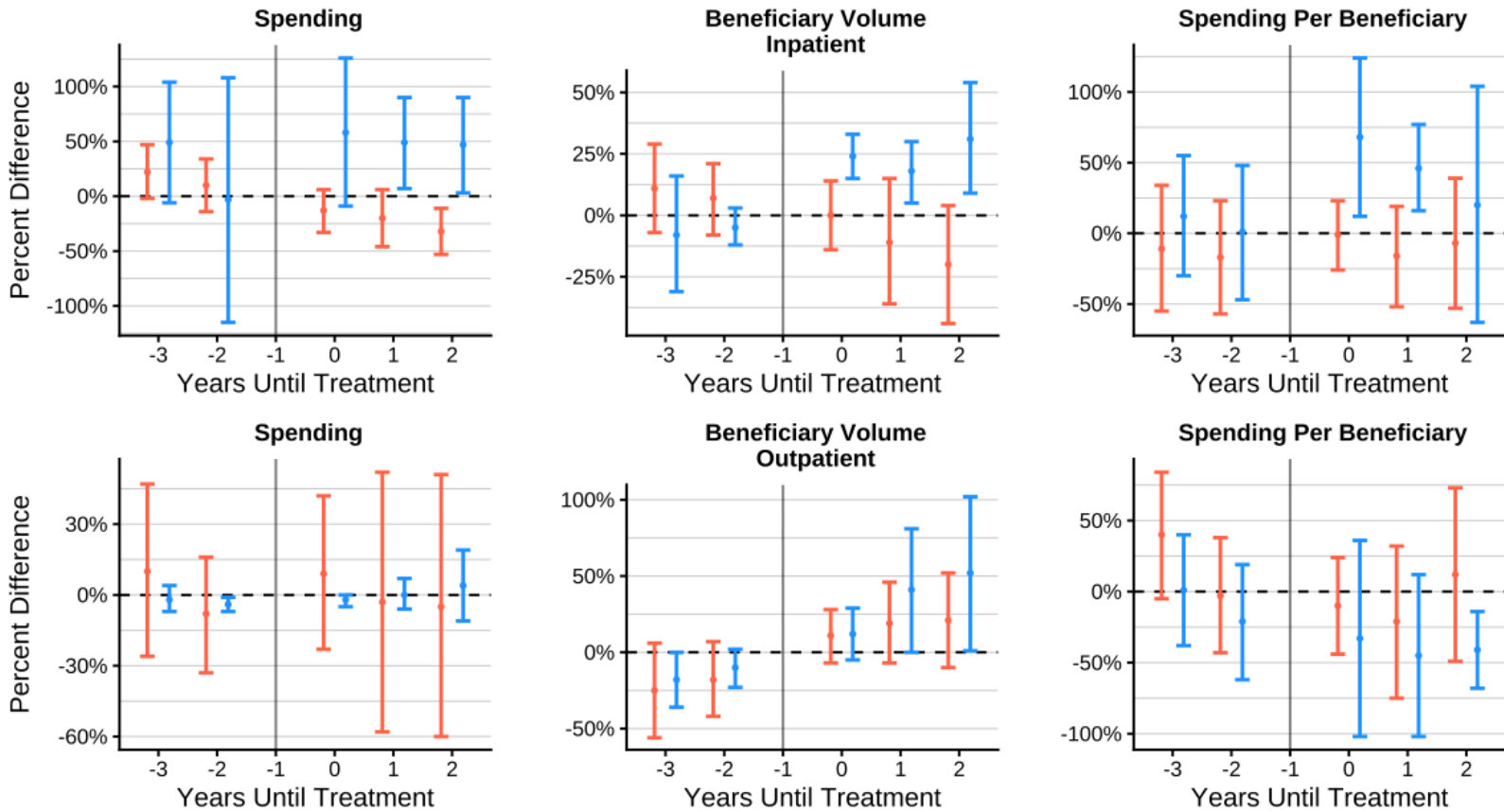
eFigure 1. Facility Average Inpatient and Outpatient Spending, Volume, and Spending Per Beneficiary in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Behavioral Health Treatment Facilities

Acquisition Status — Never Acquired — PE Acquired — Other For-Profit Acquired

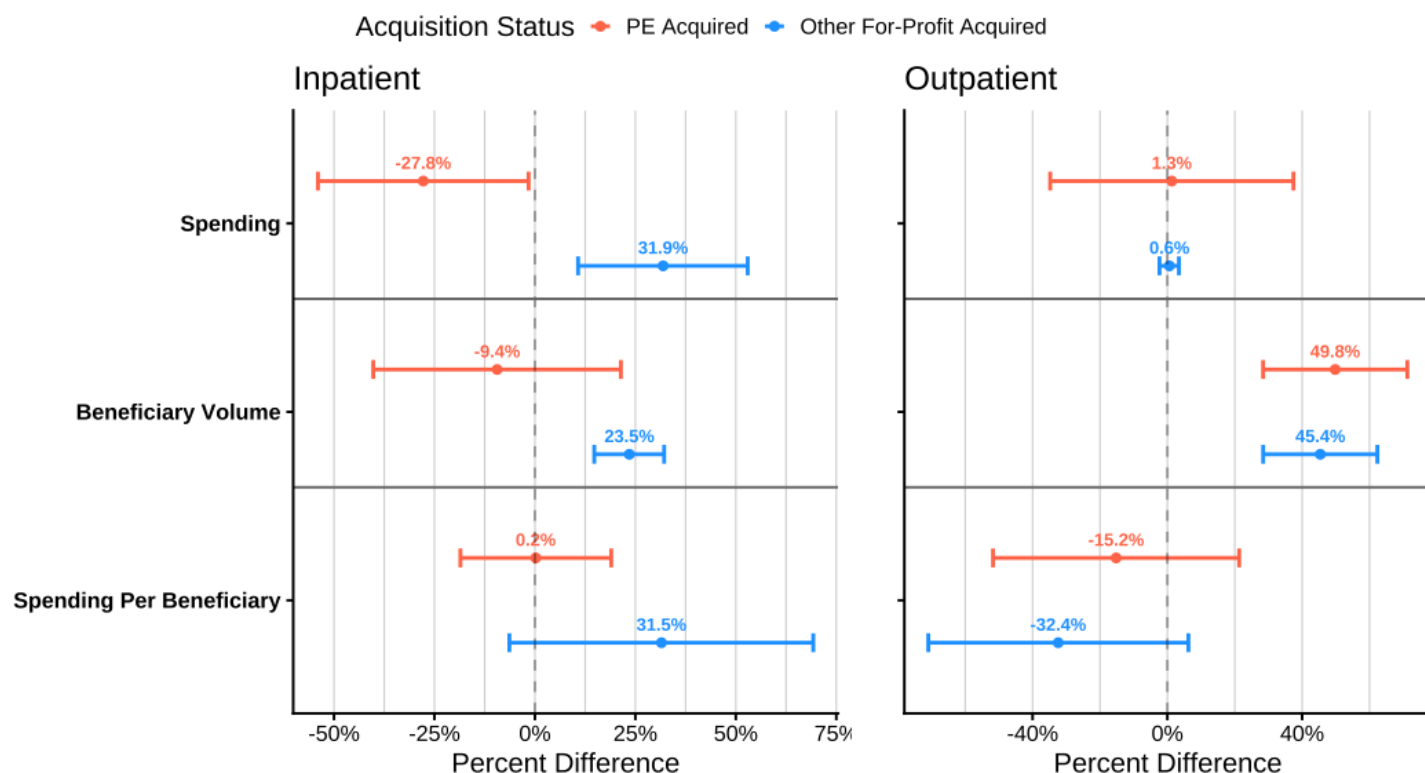


eFigure 2. Event Studies of Acquisition on Percent Change Inpatient and Outpatient Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type

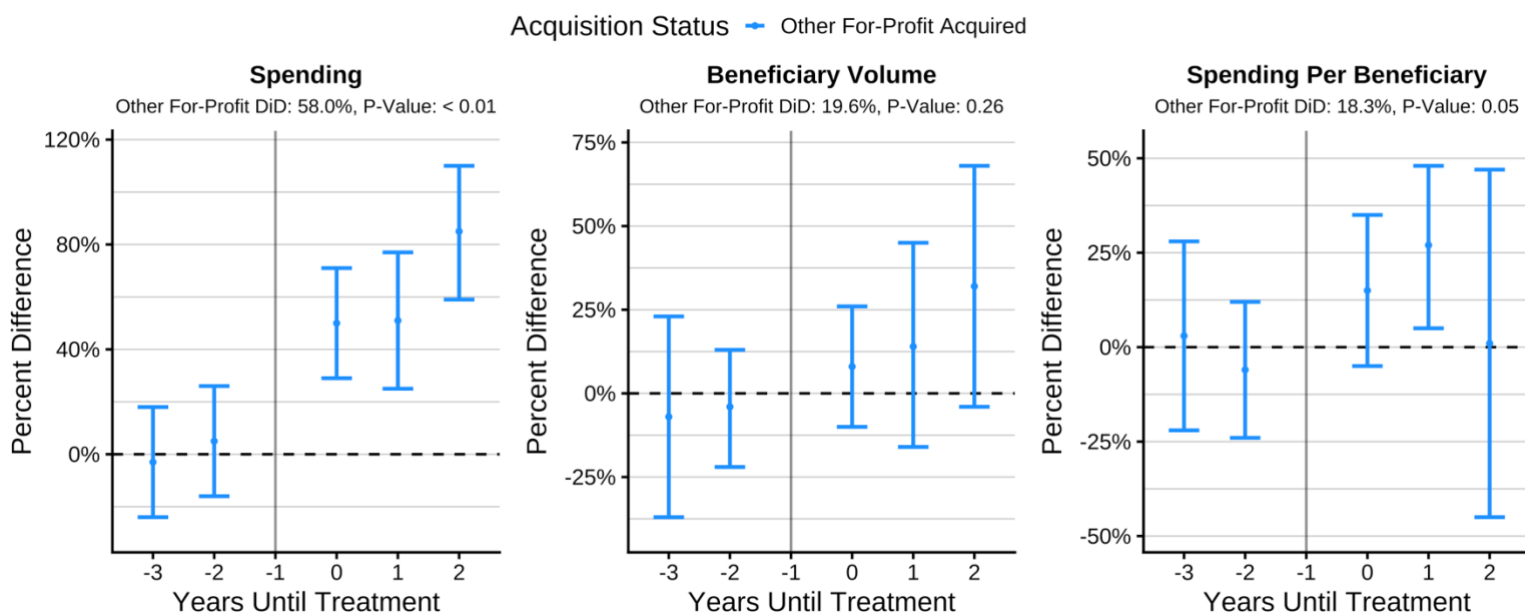
Acquisition Status PE Acquired Other For-Profit Acquired



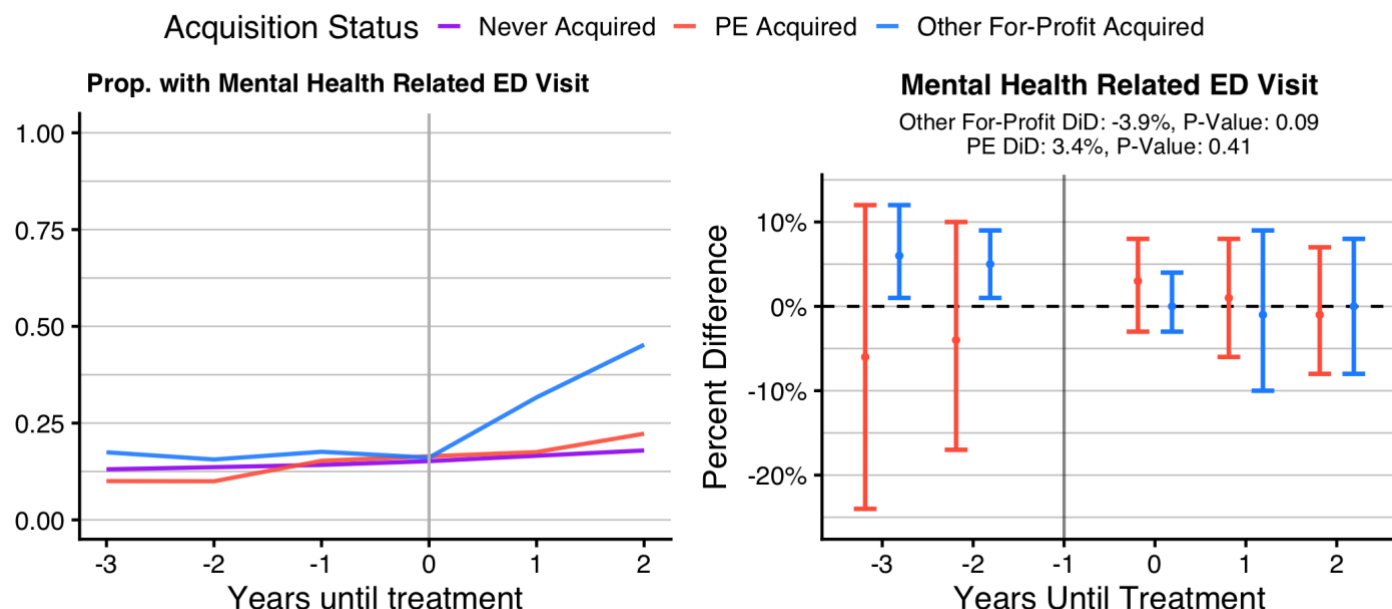
eFigure 3. Average Change in Inpatient and Outpatient Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type



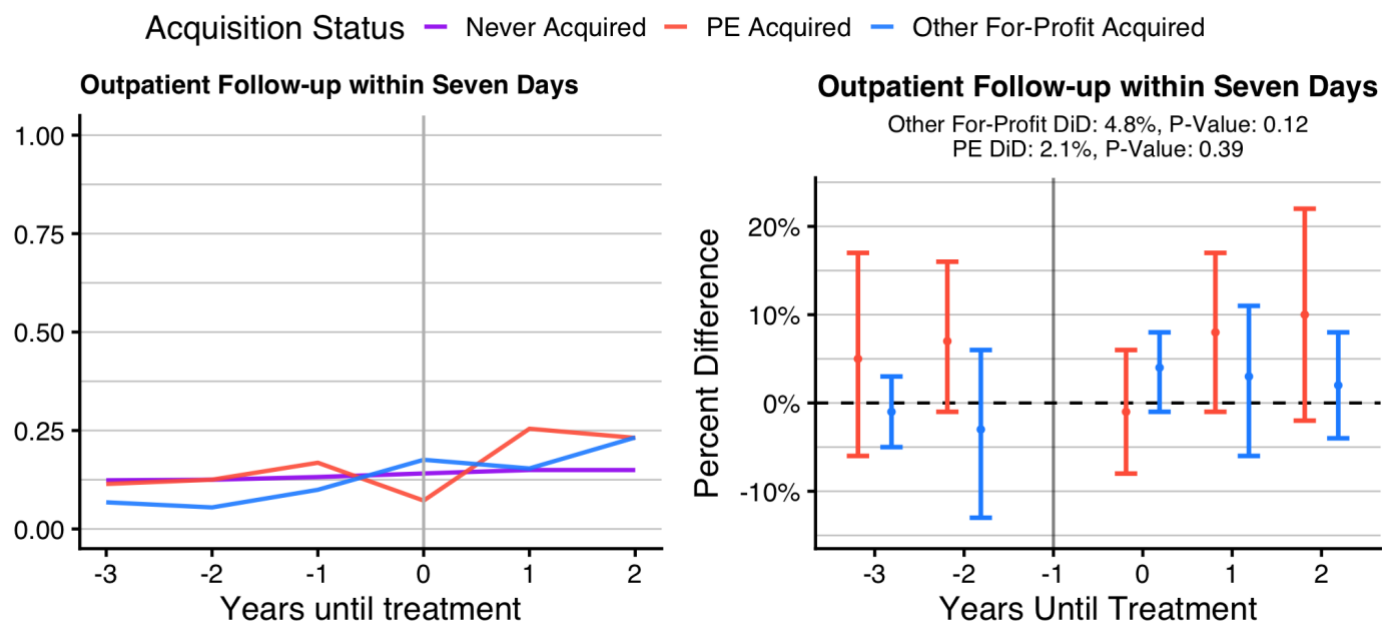
eFigure 4. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, Among Other For-Profit Relative to PE Acquired Facilities



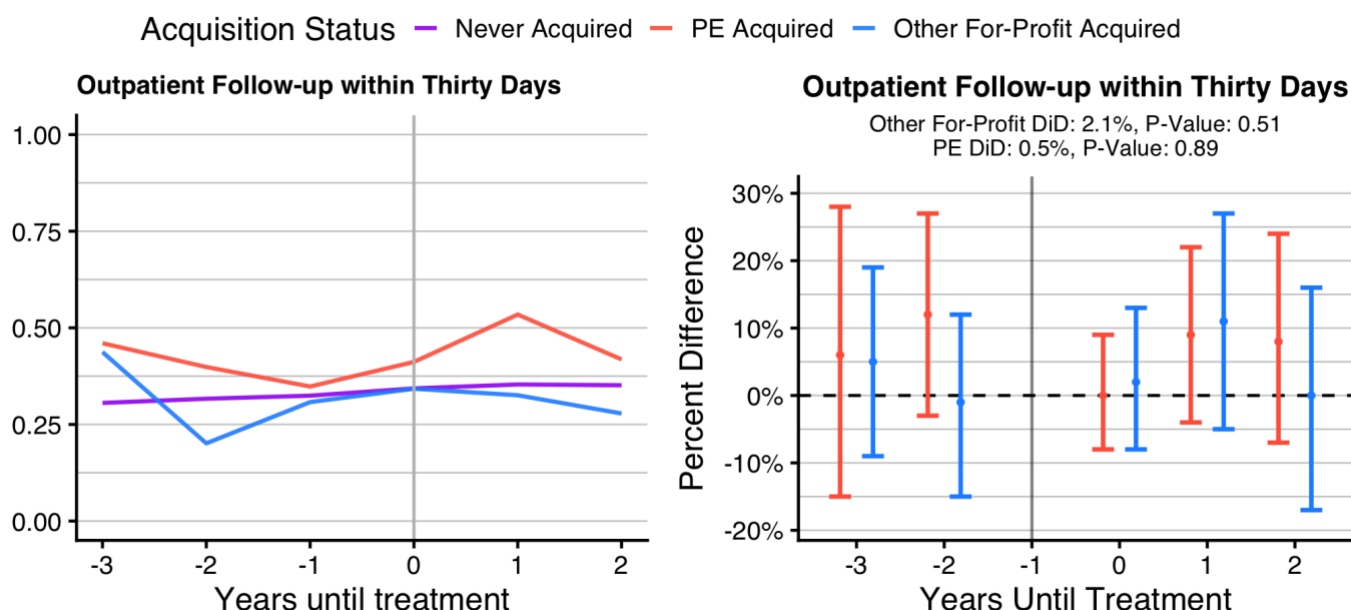
eFigure 5. Proportion of Beneficiaries That Had a Mental Health-Related Emergency Department Visit in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



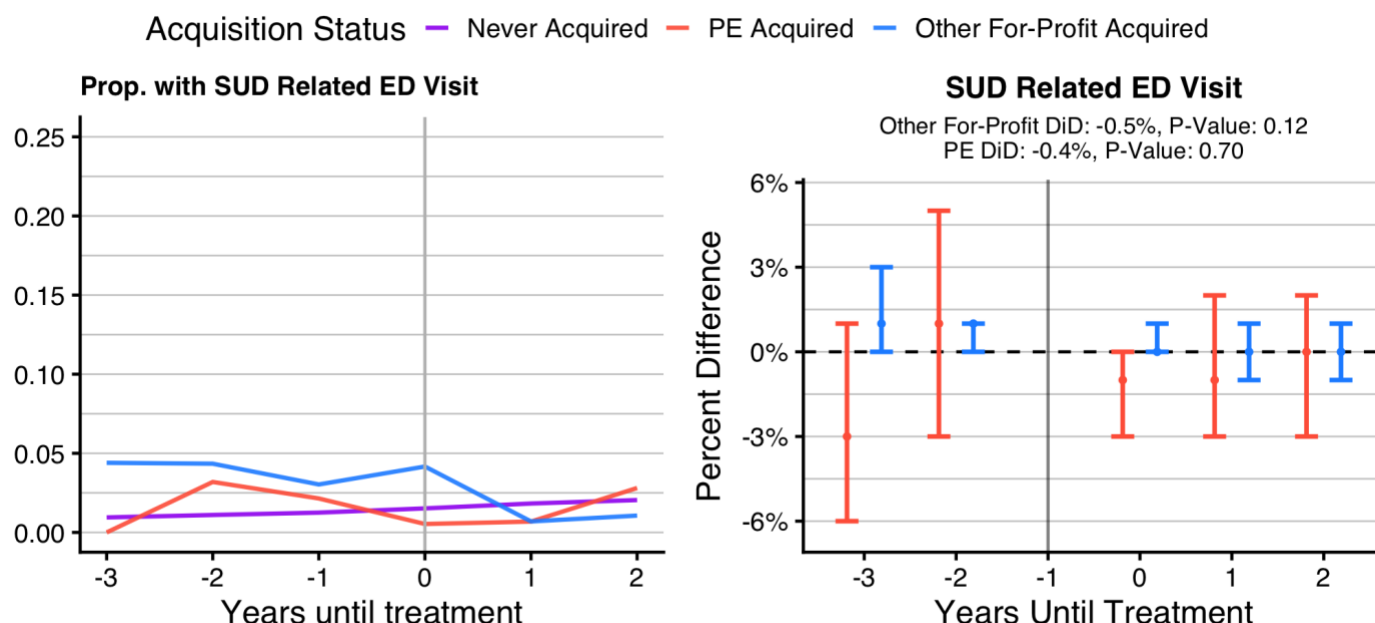
eFigure 6. Proportion of Beneficiaries That Received Outpatient Follow-Up Care Within Seven Days of Having a Mental Health-Related Emergency Department Visit in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



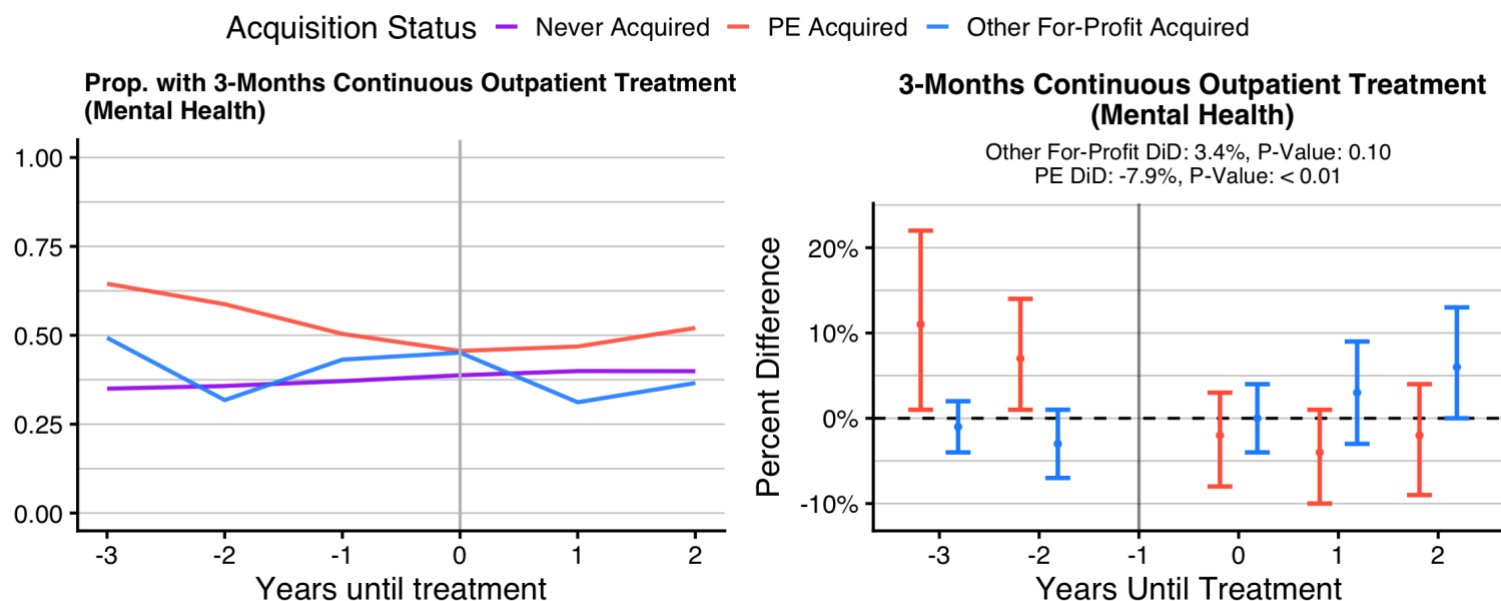
eFigure 7. Proportion of Beneficiaries That Received Outpatient Follow-Up Care Within Thirty Days of Having a Mental Health-Related Emergency Department Visit in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



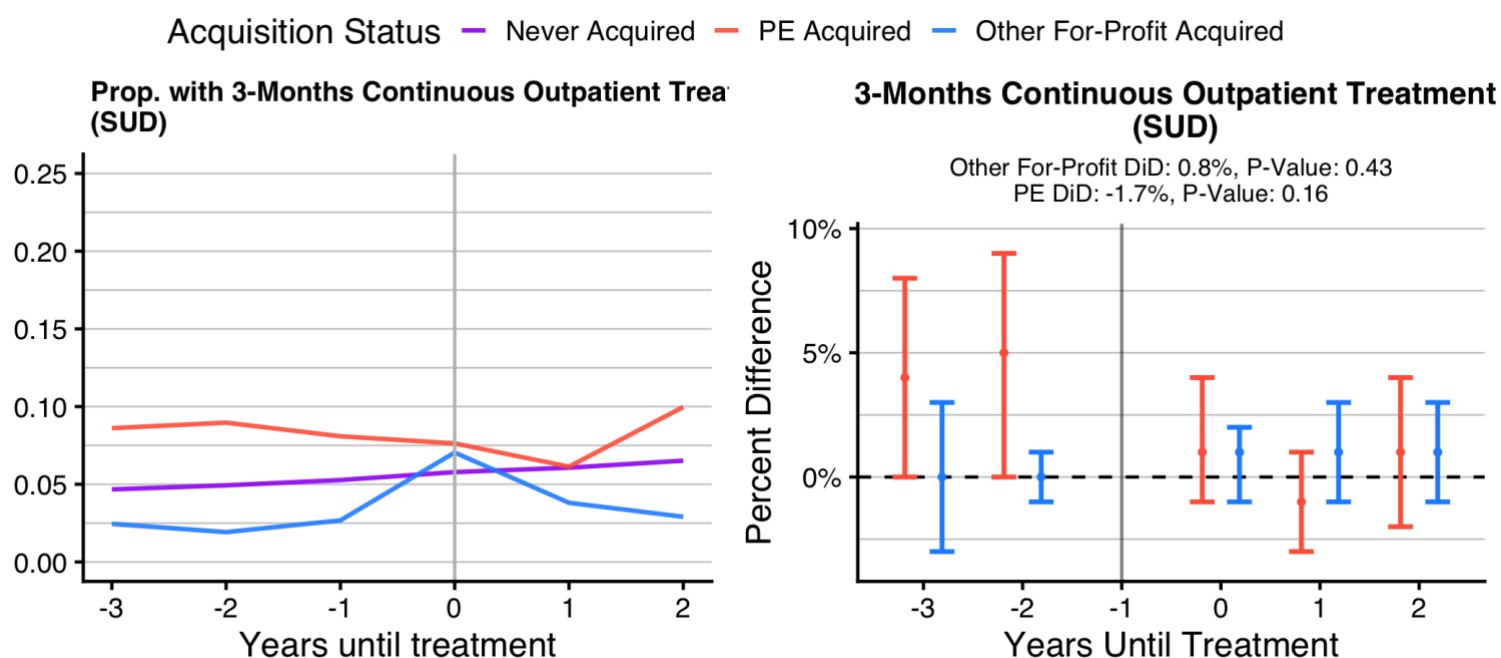
eFigure 8. Proportion of Beneficiaries That Had an SUD-Related Emergency Department Visit in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



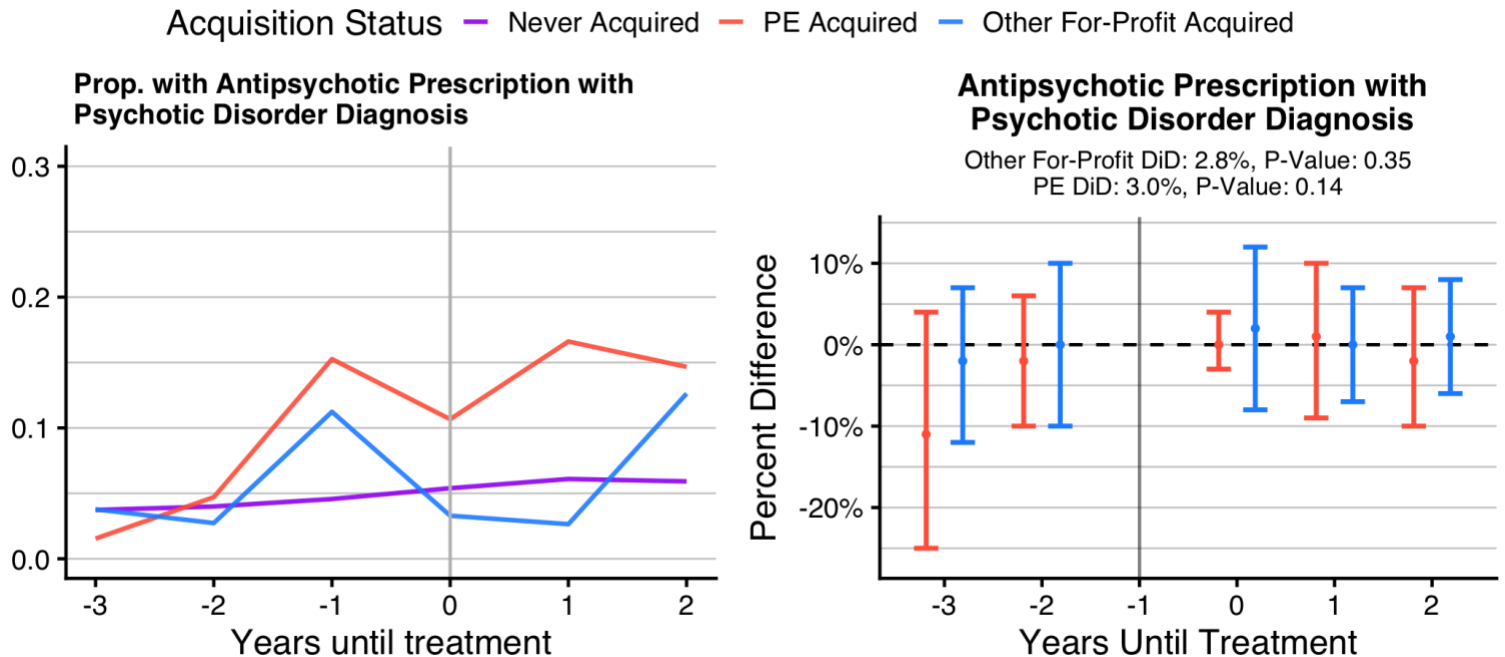
eFigure 9. Proportion of Beneficiaries Receiving at Least Three Months of Continuous Outpatient Mental Health Treatment in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



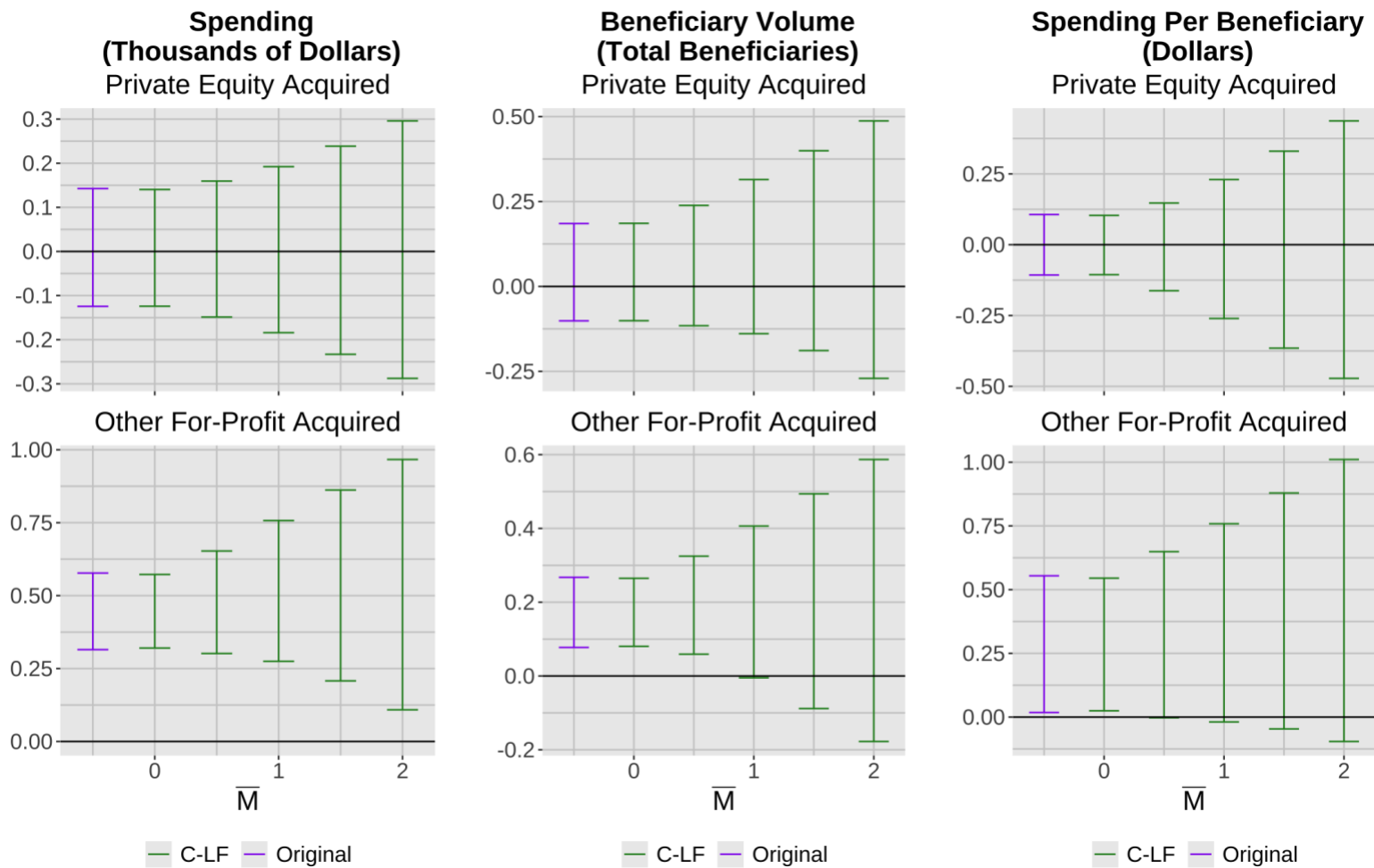
eFigure 10. Proportion of Beneficiaries Receiving at Least Three Months of Continuous Outpatient SUD Treatment in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



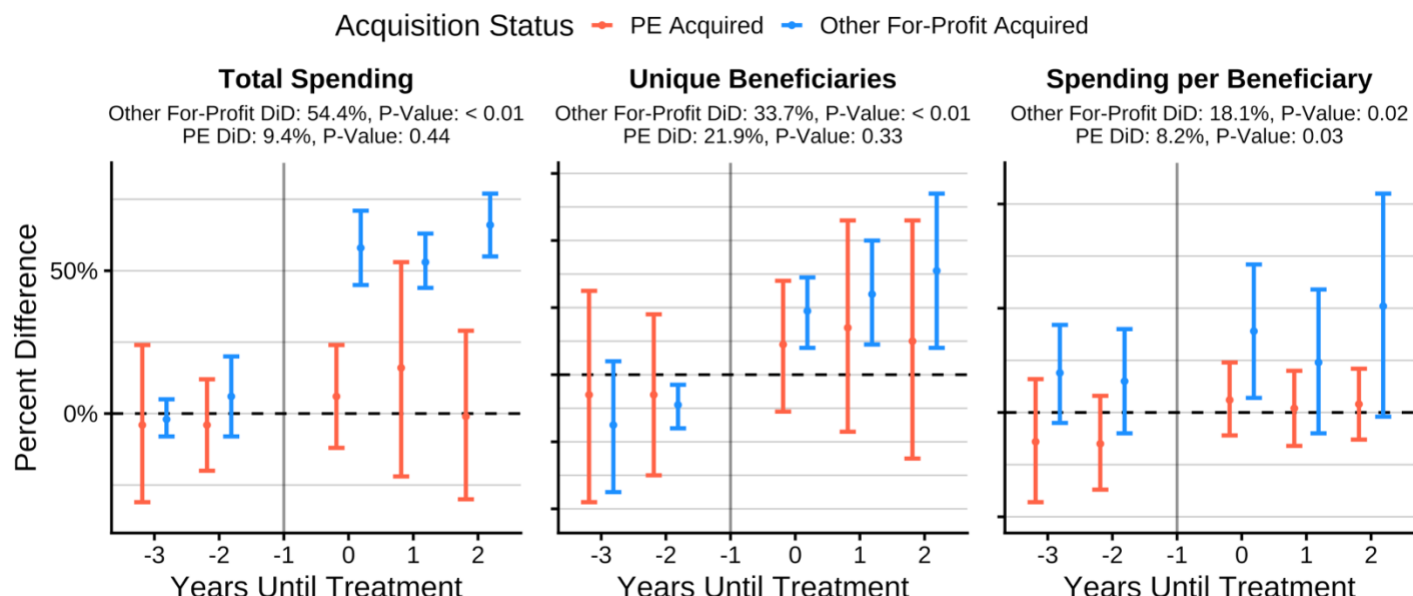
eFigure 11. Proportion of Beneficiaries With a Psychotic Disorder that Received an Antipsychotic Prescription in Years Relative to Acquisition Among PE, Other For-Profit, and Never Acquired Facilities, and their Corresponding Event Studies and Difference-in-Differences Estimates



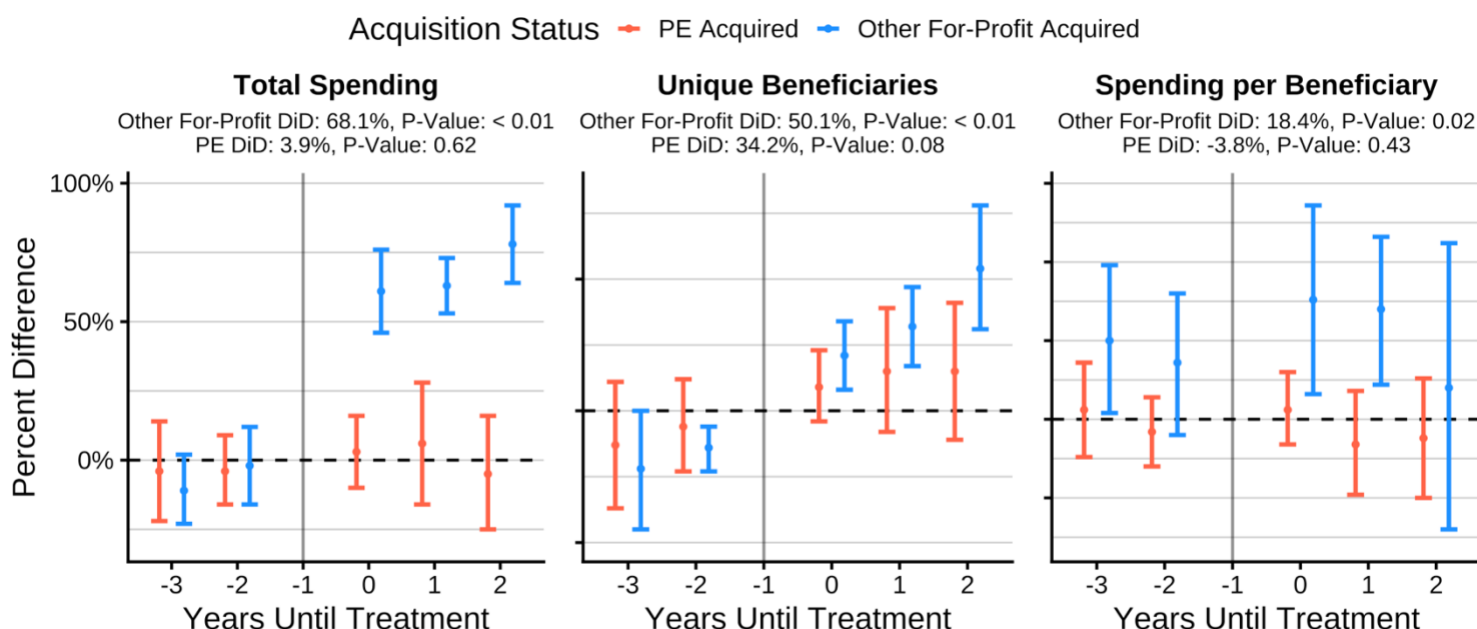
eFigure 12. Honest Difference-in-Differences Event Studies of Acquisition on Percent Change in Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type and Service Type



eFigure 13. Event Studies and Corresponding Difference-in-Differences Estimates of Acquisition on Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type and Service Type, Omitting Facilities Offering Highly Specialized Treatment (Opioid Treatment Programs, Alzheimer's/Memory Care, Eating Disorder Clinics)

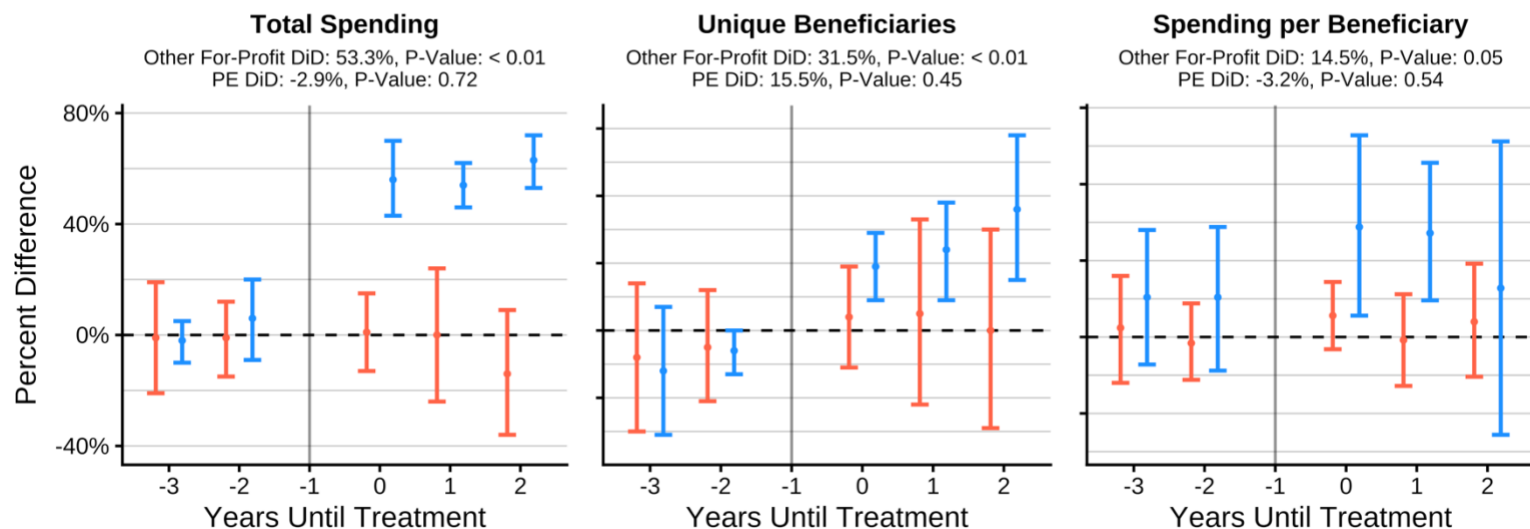


eFigure 14. Event Studies and Corresponding Difference-in-Differences Estimates of Acquisition on Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type and Service Type with Additional Matching on Facility Ownership (Private For Profit, Private Nonprofit, Government Owned)



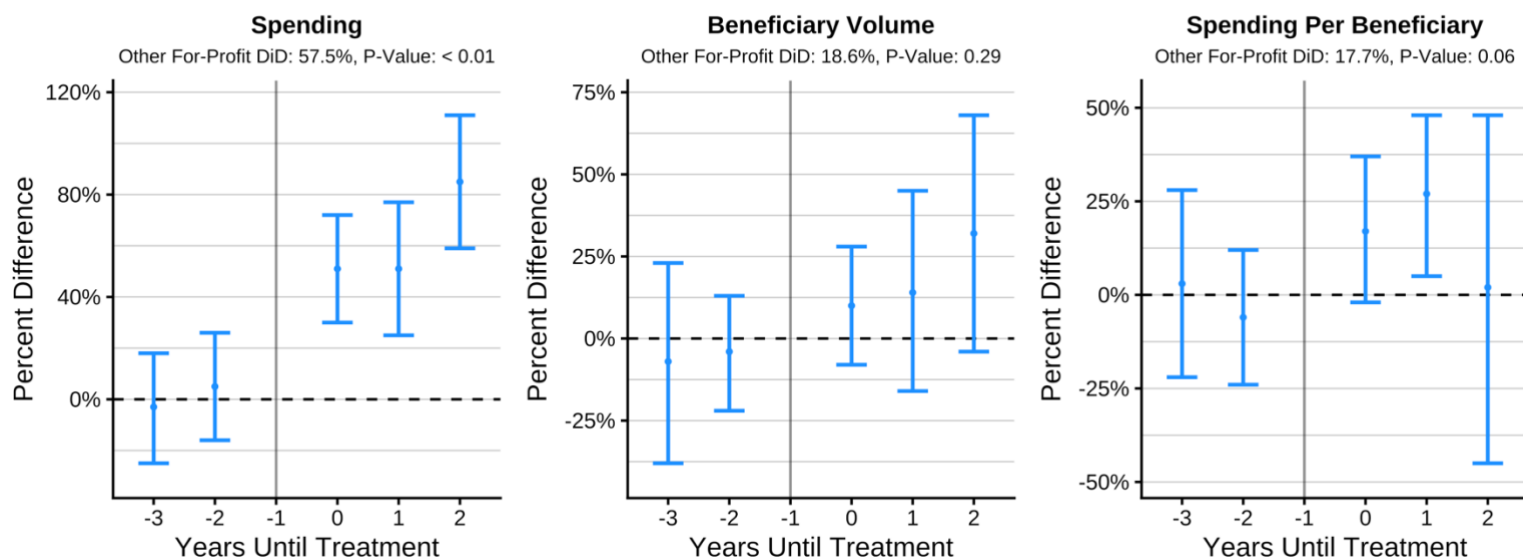
eFigure 15. Event Studies and Corresponding Difference-in-Differences Estimates of Acquisition on Total Beneficiary Spending, Volume, and Spending Per Beneficiary by Acquirer Type and Service Type, with State Fixed-Effects

Acquisition Status — PE Acquired — Other For-Profit Acquired

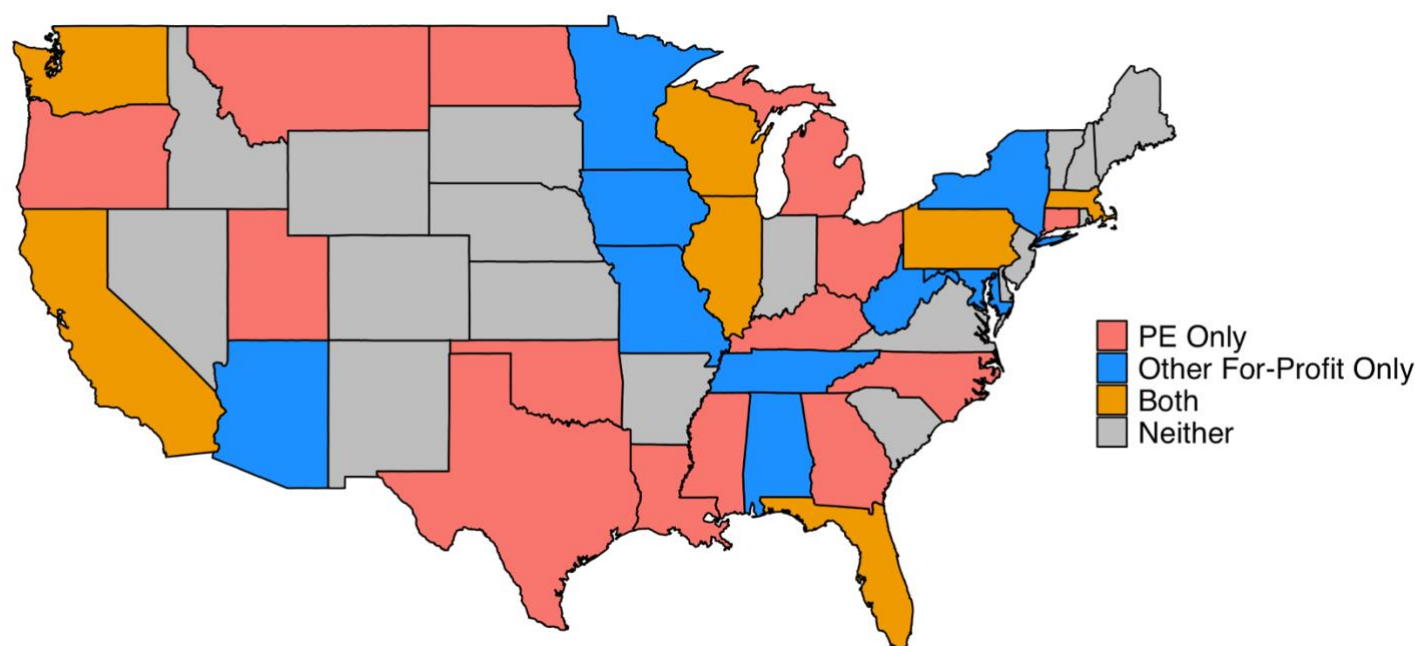


eFigure 16. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, Among Other

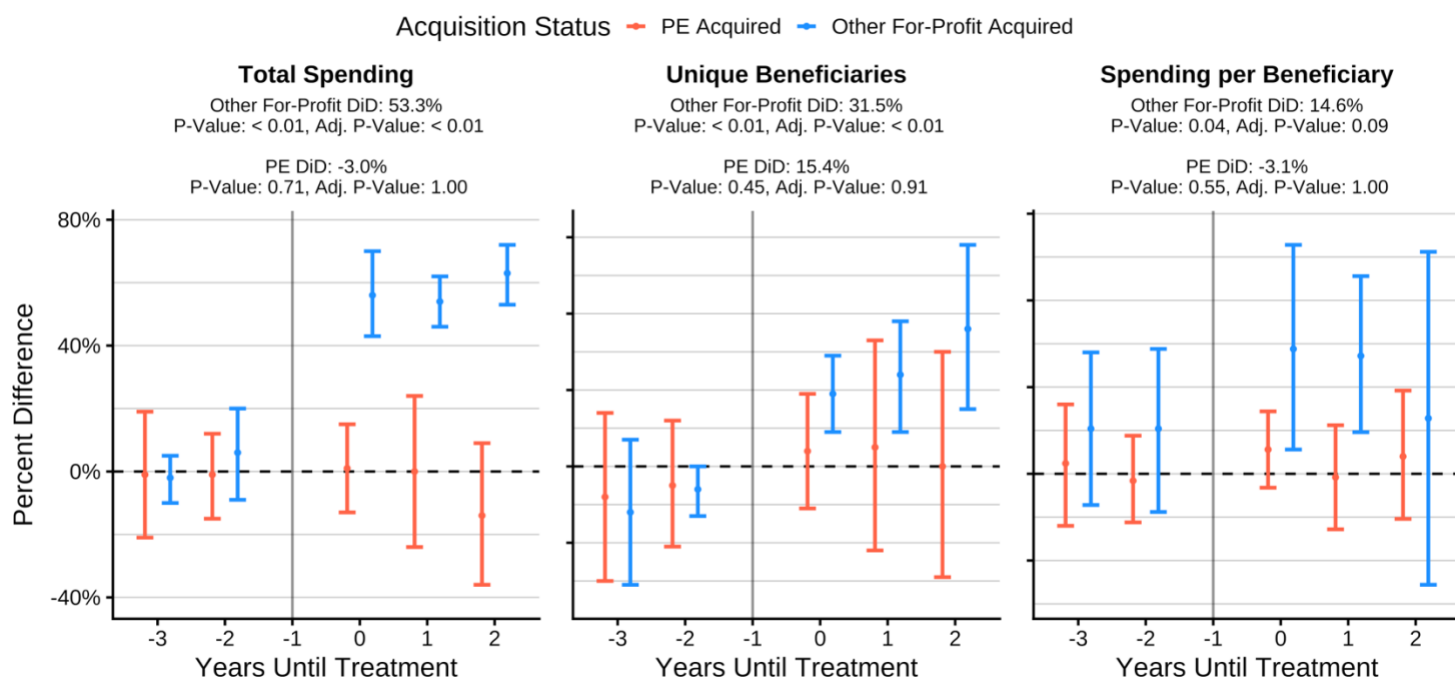
Acquisition Status — Other For-Profit Acquired



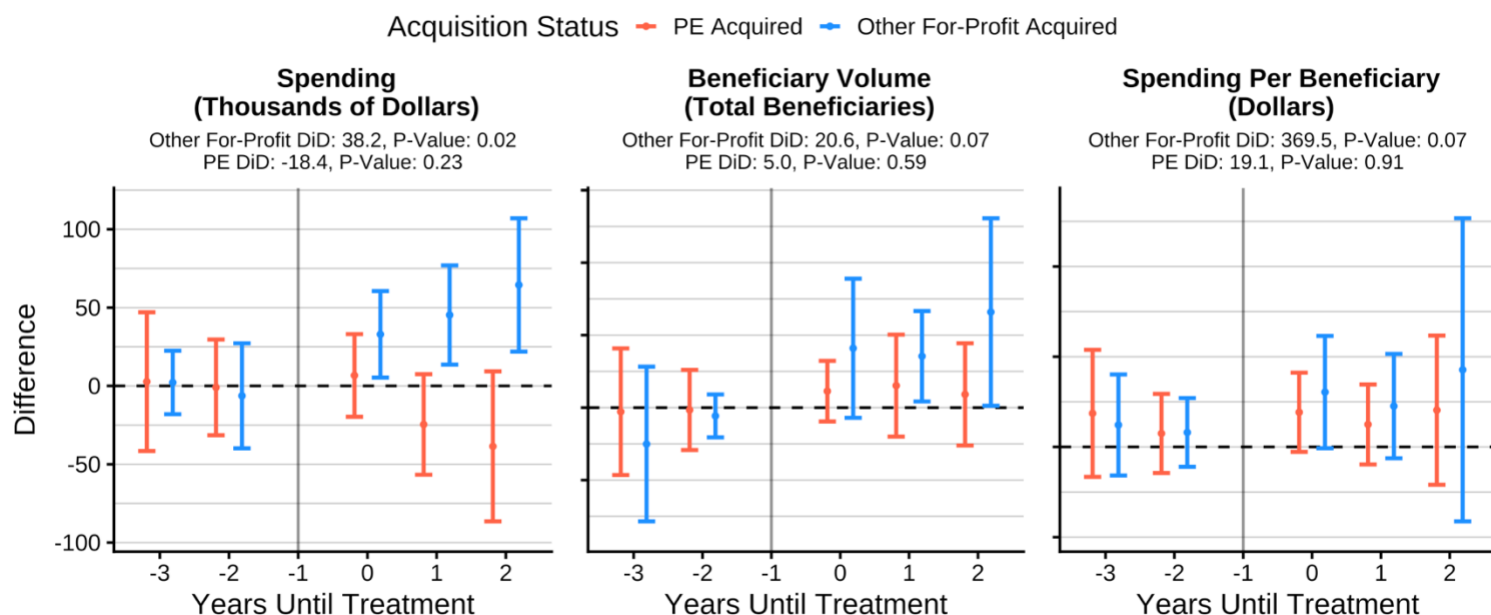
eFigure 17. Distribution of Acquisitions by Whether a State Experienced an Acquisition by PE, Another For-Profit Firm, Both, or Neither, Between 2014 and 2022



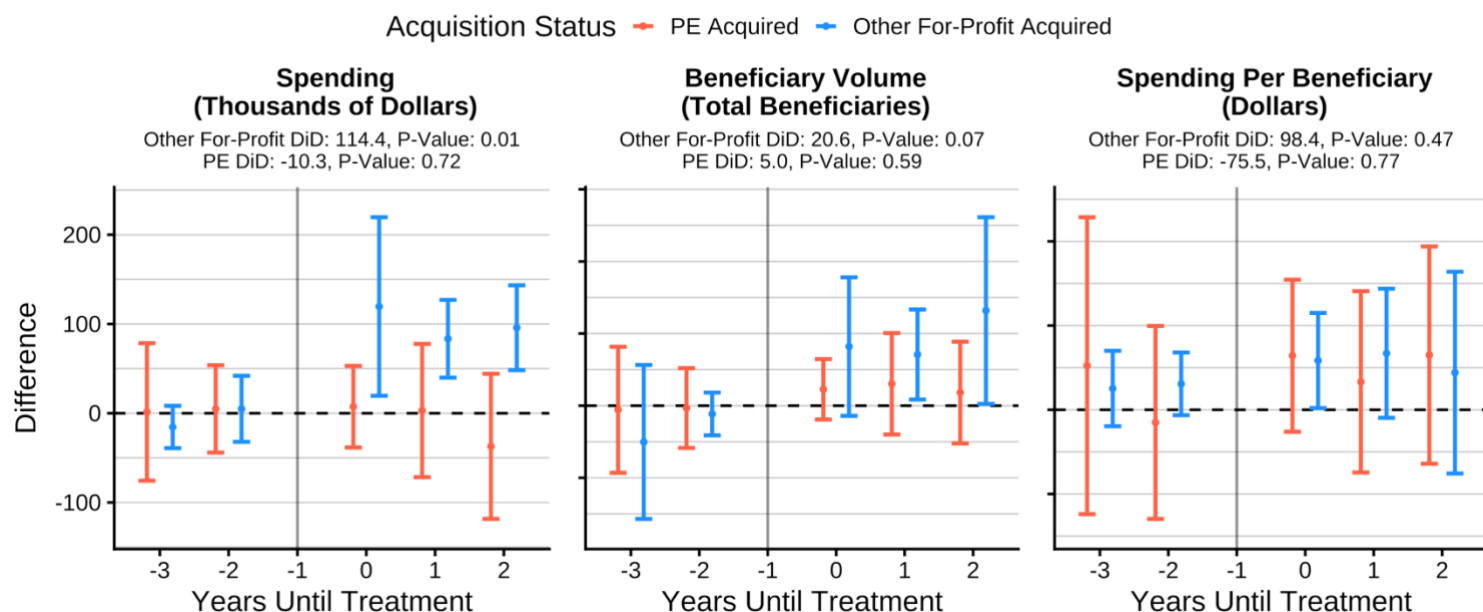
eFigure 18. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, with and without Bonferroni Adjustment



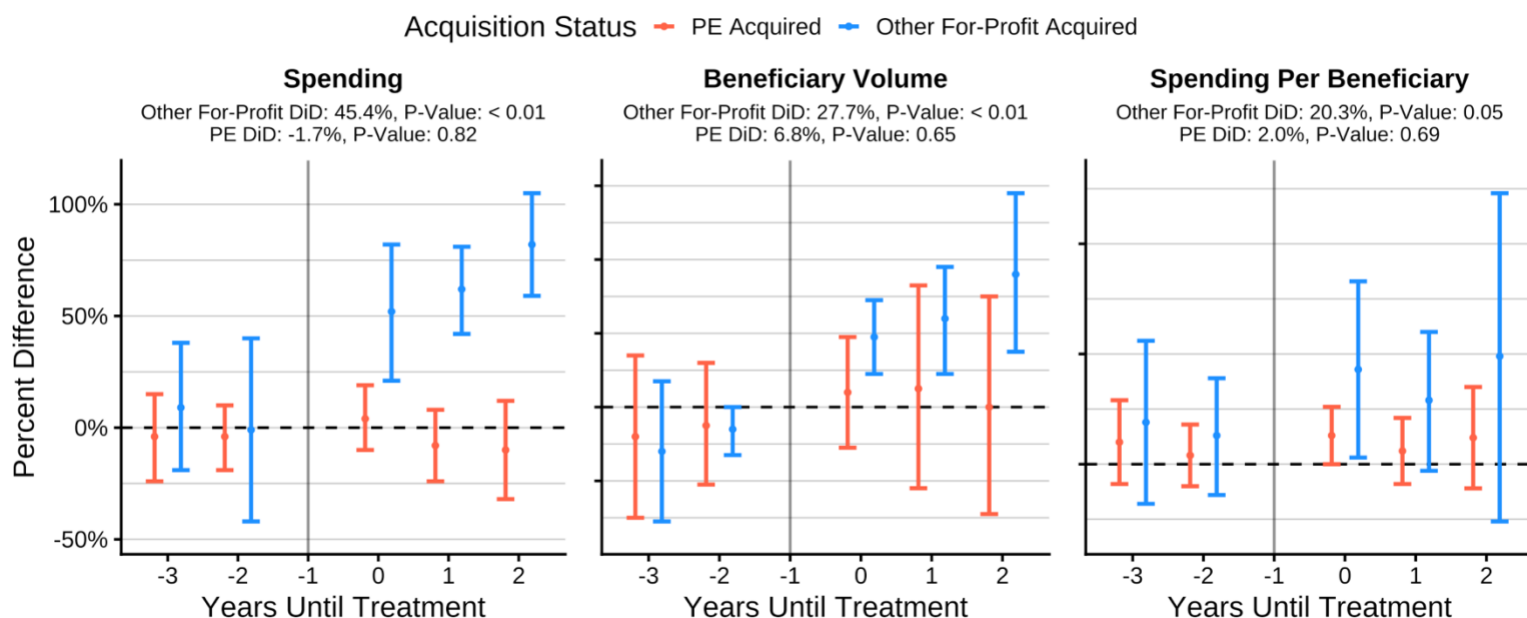
eFigure 19. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, Estimated via Facility Volume-Weighted Least Squares



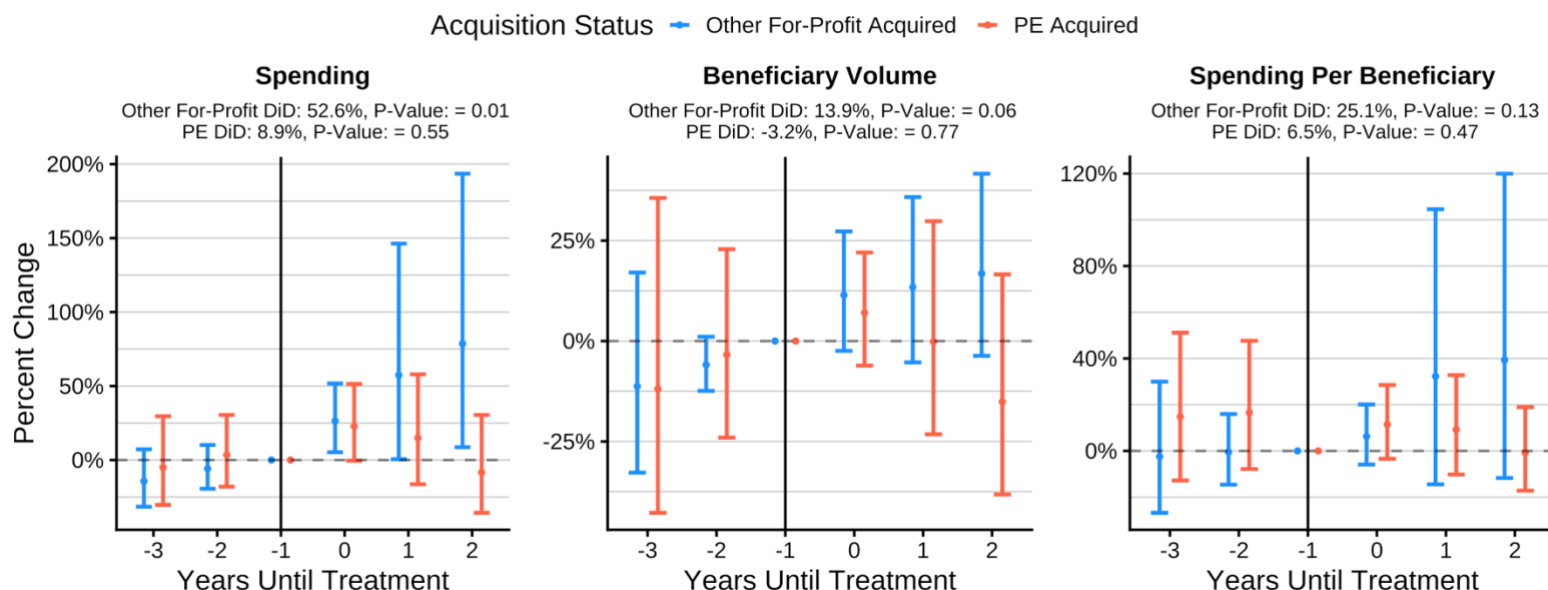
eFigure 20. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, Estimated via Ordinary Least Squares



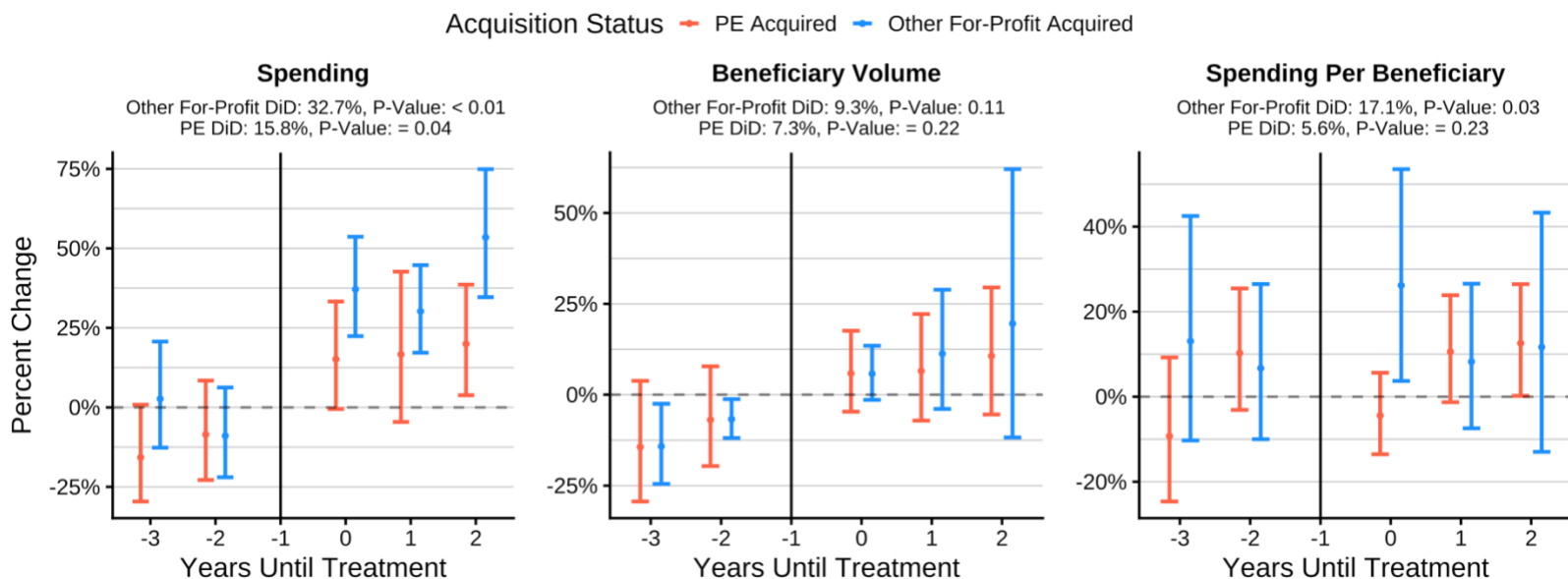
eFigure 21. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, Estimated via Unweighted Poisson Regression



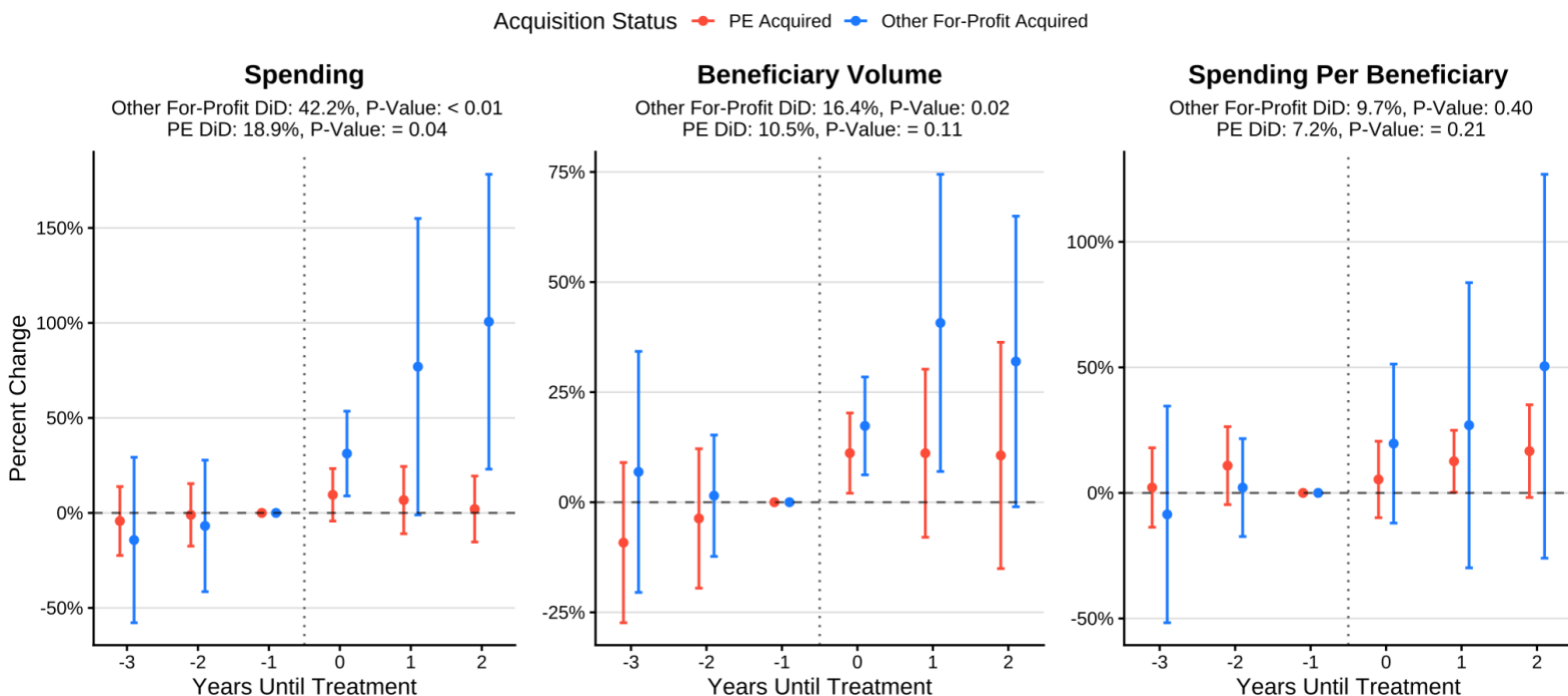
eFigure 22. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary (Callaway and Sant'Anna)



eFigure 23. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary (Sun and Abraham)



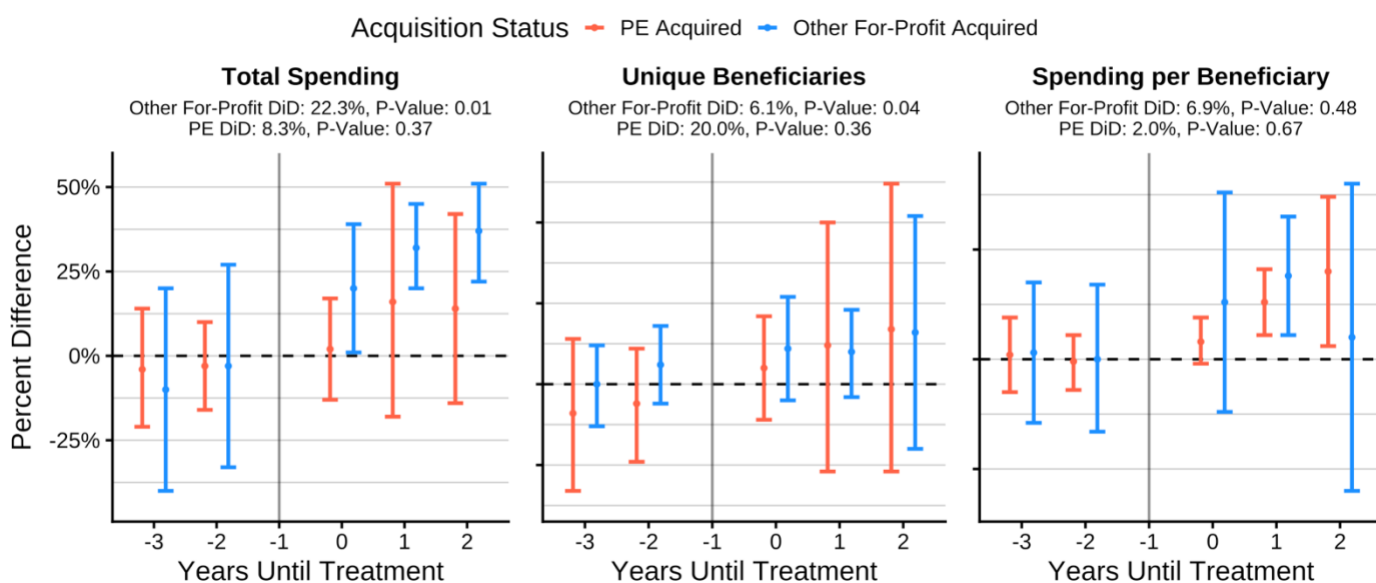
eFigure 24. Event Studies and Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary (Wooldridge)



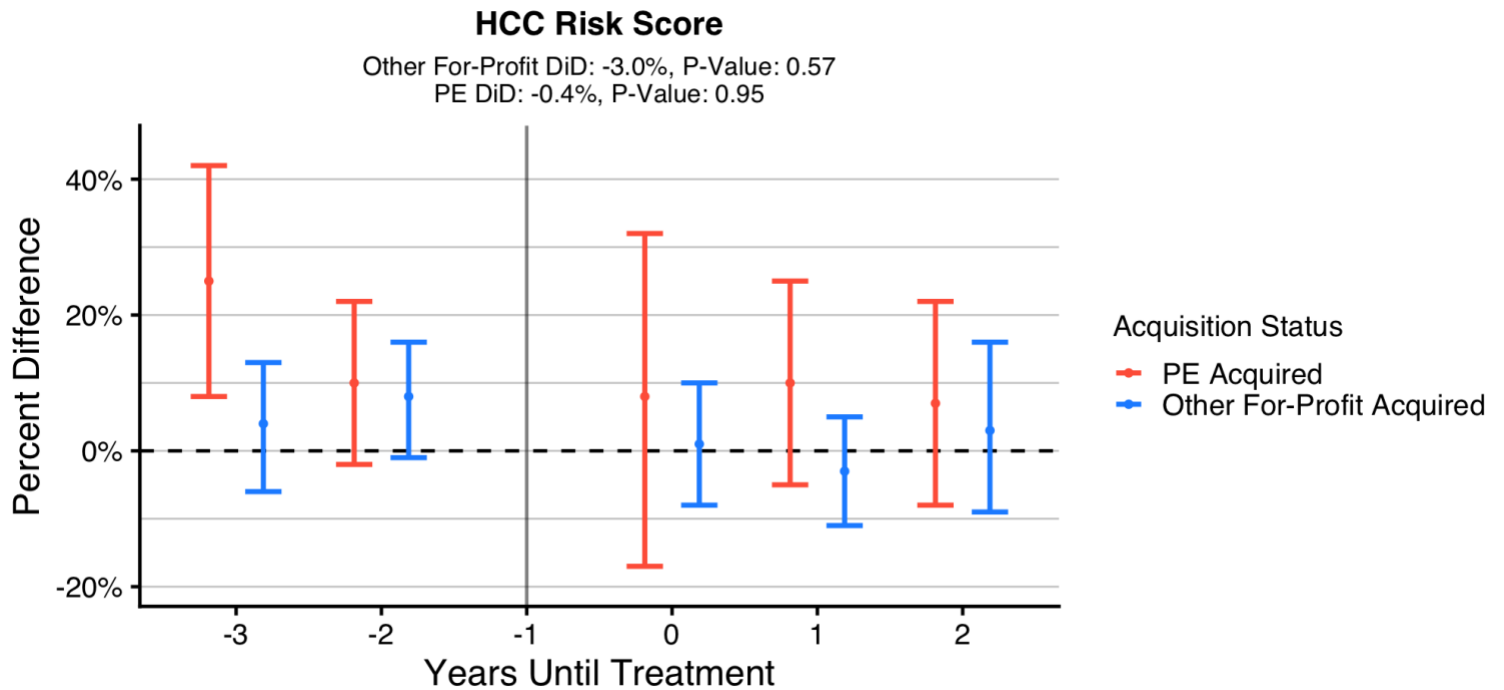
eTable 5. Difference-in-Differences Estimates of Spillovers to Non-Acquired Facilities Following an Acquisition in Their County

Outcome	DiD	95% Confidence Interval		Std. Error	P-Value
Claim Volume	1.9%	2.1%	-2.3%	6.2%	0.39
Spending	-0.7%	0.7%	-2.0%	0.6%	0.28
Outpatient Spending	1.7%	1.8%	-1.8%	5.3%	0.35
Inpatient Spending	-0.7%	0.7%	-2.1%	0.7%	0.34
Spending per Bene	-1.1%	1.5%	-3.9%	1.8%	0.46
Total Beneficiary Volume	-0.2%	1.5%	-3.0%	2.8%	0.91
Attributed Beneficiary Volume	1.4%	1.3%	-1.2%	4.1%	0.30
Outpatient Volume	-0.6%	2.3%	-5.1%	4.0%	0.78
Inpatient Volume	0.2%	1.8%	-3.2%	3.8%	0.89
HCC Risk Score	-0.2%	0.4%	-0.9%	0.5%	0.58
SUD-Related ED Visits per Bene	21.4%	7.9%	4.6%	40.9%	0.01
Mental Health-Related ED Visits per Bene	-8.6%	13.2%	-28.3%	16.5%	0.47
Share Benes with SMI	0.6%	0.6%	-0.6%	1.8%	0.31
Share Benes with Non-SMI MH Condition	-0.6%	0.7%	-1.9%	0.8%	0.42
Share of Benes with SUD	1.1%	0.9%	-0.6%	2.9%	0.20
Share Dual Enrolled	0.3%	0.5%	-0.8%	1.4%	0.58
Share with 3 Months Treatment Continuity	-0.8%	1.3%	-3.3%	1.8%	0.54

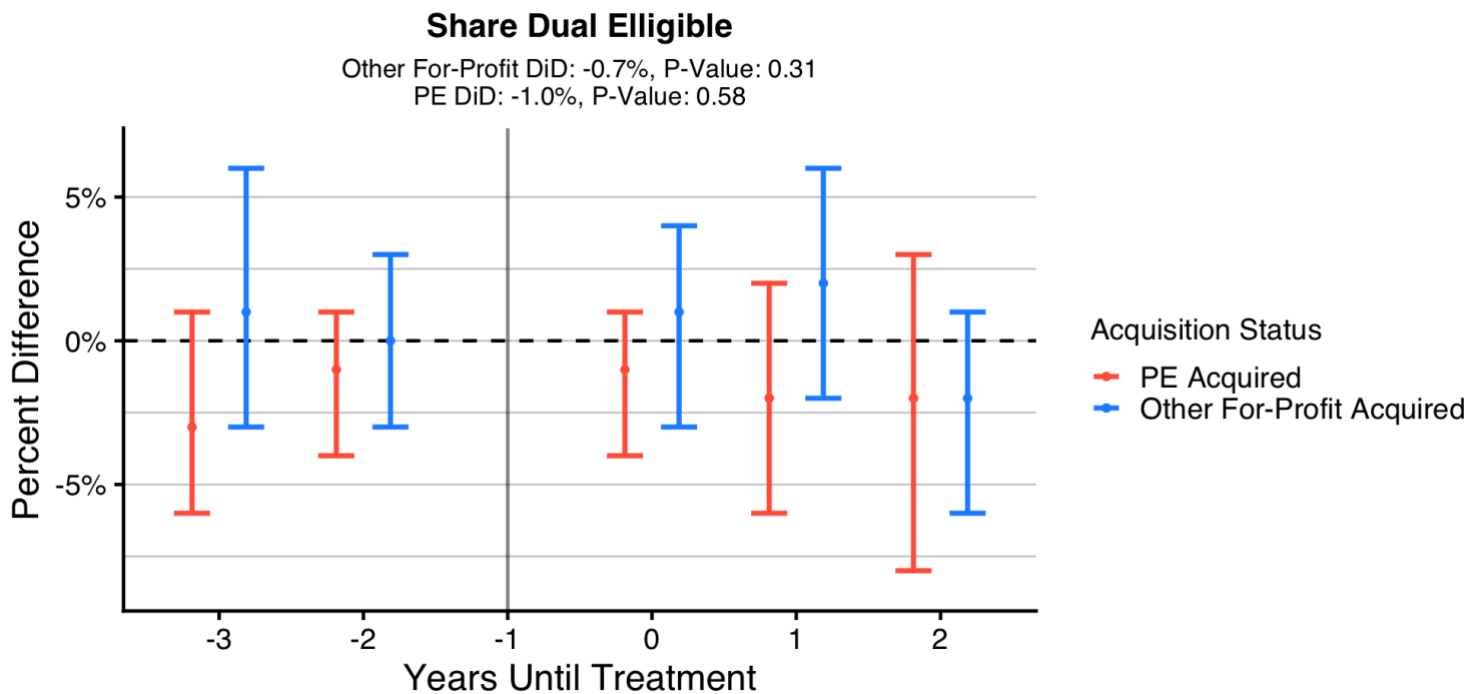
eFigure 25. Event Studies and Their Corresponding Difference-in-Differences Estimates of Acquisition on Beneficiary Spending, Volume, and Spending Per Beneficiary, using 2014-2019 Data



eFigure 26. Event Studies and Their Corresponding Difference-in-Differences Estimates of Acquisition on Beneficiary HCC Risk Score by Acquirer Type

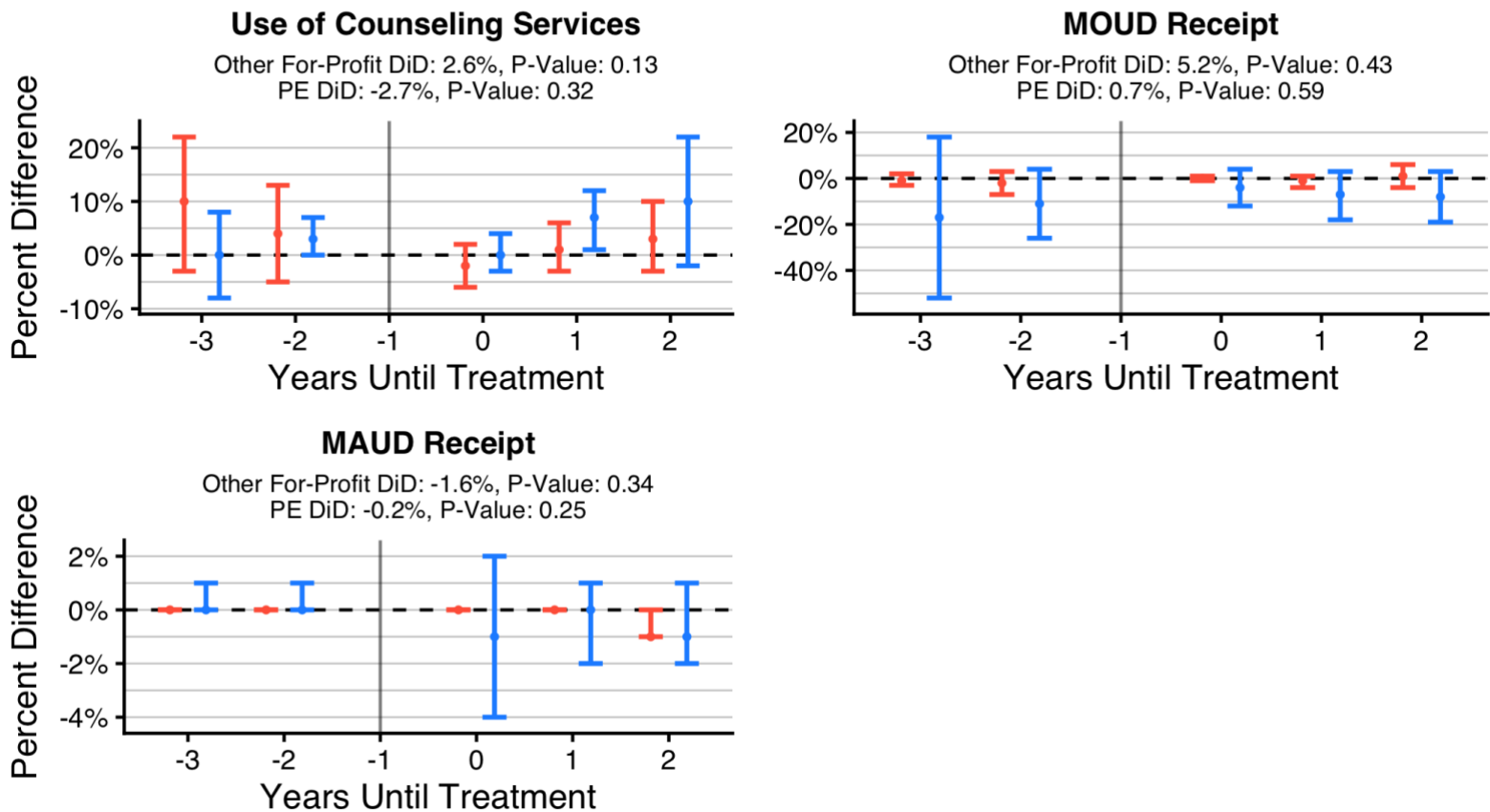


eFigure 27. Event Studies and Their Corresponding Difference-in-Differences Estimates of Acquisition on the Proportion of Beneficiaries that are Dually Eligible by Acquirer Type

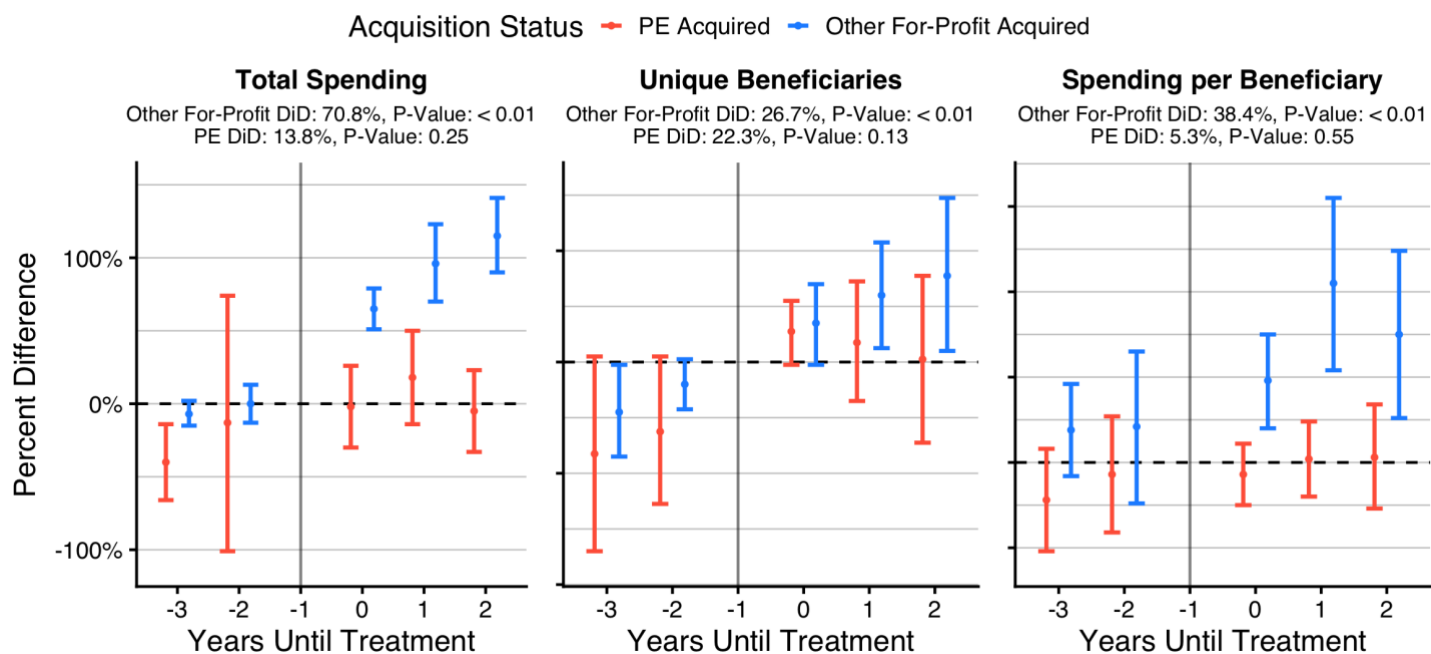


eFigure 28. Event Studies and Their Corresponding Difference-in-Differences Estimates of Acquisition on the Proportions of Beneficiaries that Used Counseling Services, Received Medications for Opioid Use Disorder (MOUD), or Received Medications for Alcohol Use Disorder, Respectively

Acquisition Status — PE Acquired — Other For-Profit Acquired



eFigure 29. Event Studies and Difference-in-Differences Estimates of Acquisition on Attributed Beneficiary Spending, Attributed Beneficiary Volume, and Spending Per Attributed Beneficiary



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